

Prospective Health under Declared Specifications

An Auditable Empirical-Synthetic Forest Measurement Study

Zed James

Paper V in the series: *Health, Formally Defined*.

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Preprint – October 09 2026.
doi:10.5281/zenodo.23268987.

Abstract

A formal prospective Health proposition combines present realization of an organization with the adequacy of its viable continuation capacity. We examine the empirical conditions required to evaluate that proposition through an auditable stochastic measurement system, using longitudinal woody-stem observations from the 35-ha Harvard Forest ForestGEO plot. The study addresses identification and calibration of latent states and demographic processes, reliability of prospective adequacy decisions, and sufficiency of the response representation for the declared query. The computational specification includes an individual-exposure reconstruction kernel, explicitly sampled demographic parameters, a mean-preserving nonnegative Gamma growth law, original-law-weighted viable subprobability capacity, a single reserve-qualified adequacy query, and distinct finite-bank and stochastic-kernel decisions.

Masking experiments on securely observed stems reveal 47–56% empirical coverage for nominal 90% individual-diameter intervals, with appreciably better aggregate structural coverage. Identity analysis establishes that none of 6,992 later-only living stem records uniquely identifies biological threshold entry. Individual-time projection changes some finite-design Health classifications and yields large local changes in continuation capacity despite small average state shifts. Independent 2021–2024 adult-mortality observations provide a support-aligned component test: among 21,576 strictly scored stems, 1,477 deaths were observed and 1,050.3 predicted; hemlock mortality accounts for 685 observed versus 280.3 predicted deaths among 7,557 stems, with a spatial-cluster descriptive interval of $[0.0456, 0.0618]$ for the mortality-rate gap. Independently observed seedling transitions supply a distinct descriptive carrier and leave whole-plot regenerative entry unidentified.

Under a known-model, full-context synthetic benchmark well separated from the decision boundary, balanced accuracy reaches 99.22%. In a boundary-concentrated design, observation-only discrimination is approximately 56%; a four-way state/parameter oracle analysis identifies strong interactions between reconstruction and parameter inference. Missingness, contextual priors, and model discrepancy alter selected decisions, while a finite counterexample proves that cohort RMS diameter can erase information required for juvenile-sensitive adequacy. All computations preserve failure mass under the original candidate measure and distinguish executed specification fidelity from empirical measurement validity. The study distinguishes exact mathematical specification, executed finite-design evidence, and observationally supported prediction. Path-level replay localizes the selected one-year exposure contrast entirely to terminal juvenile-support failure (4,053 versus 137 failed histories in two 4,096-path banks), with no basal-area or model-lawfulness failures. Under the fitted shared-cell-hazard predictive law, 685 observed hemlock deaths lie beyond its central 95% predictive interval of 220–347 deaths ($\Pr\{D \geq 685\} = 4.63 \times 10^{-21}$), while the corresponding nonhemlock count lies within its predictive distribution. These findings distinguish structural threshold sensitivity from model predictive miscalibration. Operational forest Health remains an unestablished empirical target.

1 Introduction

A measurement procedure for prospective Health must connect observed data to a defined organization, a distribution of potential continuations, a response representation, and a requirement evaluated on those responses. The governing proposition is

$$\boxed{\text{Health}_{\mathfrak{S},r}(x; d, t) \iff \text{Realizes}_{\mathfrak{S}}(x) \wedge \text{Adequate}_{\mathfrak{S},r}(\text{Capacity}_{\mathfrak{S}}(x; d, t))}. \quad (1)$$

The formal program specifies that proposition and its contextual and representational requirements [1, 2, 3, 4]. Empirical implementation requires additional identification of the state from evidence, a warranted law of prospective change, and a numerical procedure sufficiently precise for the adequacy decision.

The Harvard Forest ForestGEO record provides a demanding measurement substrate: geographically defined forest support, repeated tagged stems, asynchronous census dates, unresolved fates, and an independently available later adult-mortality survey. We study a declared structural-demographic organization based on basal area and 1–10 cm stem support. These variables define an inspectable prototype question; the scientific validity of their constitutive interpretation remains an empirical responsibility. Candidate trajectories are generated conditionally on fitted hazard and growth distributions and stipulated intervention modifiers. Their predictions are evaluated with exact accounting and controlled observation experiments.

Three questions organize the study. *Identification and calibration*: which state and demographic quantities are supported by the observed records, which rely on reconstruction, and which are unidentified? *Decision reliability*: how do state reconstruction, parameter inference, process specification, context, and simulation error affect adequacy decisions, especially near their threshold? *Query-relative representation*: does the measured state retain the distinctions required to evaluate the chosen response query? These questions supply distinct success criteria. Correct execution of an explicit specification establishes computational fidelity; a scientifically reliable measurement also requires identified inputs, calibrated response laws, adequate information and decision-relevant numerical resolution.

The work contributes an explicit end-to-end stochastic kernel, an auditable census-to-continuation probability ledger, partial observed-data calibration, an independent adult-mortality component evaluation, controlled stagewise oracle comparisons, and an exact representation-insufficiency witness. The analysis documents both performance and failure modes with their observational and computational supports. Its empirical-synthetic target is a declared finite-design Health evaluation, with full ecological Health remaining outside the validated scope.

2 The empirical object, observations, and provenance

2.1 ForestGEO object and observation windows

The target is the Harvard Forest CTFS–ForestGEO mapped plot on Prospect Hill, a 700-m by 500-m rectangle containing 875 20-m quadrats. The living-stem inclusion criterion is DBH at least 1 cm; standing dead stems at the initial census had a different inclusion convention. The WGS84 four-corner boundary is recorded in the HFD-01 object specification. The plot-local stem g_x, g_y

Observation or source class	E0	E1	Meaning in this study
Stem rows	116,227	123,218	Full recorded stem-row census across field statuses
Recorded E0 living stems	108,632	—	Baseline source for structural denominators
E1 literal <code>alive</code>	—	61,217	Conditionally interpretable dated endpoint
E1 literal <code>stem dead</code>	—	24,370	Secure subset additionally requires code corroboration
E1 literal <code>stem_gone</code>	—	37,577	Unresolved fate; mortality status unidentified
E1 literal <code>missing/broken below</code>	—	31/23	Distinct field statuses retained
E1 DBH missing	—	61,985	Missing measurement, reconstructed conditionally
E1-only measured living IDs	—	6,992	Candidate threshold entrants, subject to identity audit

Table 1: The empirical substrate. Status names reproduce the archived source labels. Stem status and tree/plant survival are separate biological variables.

transform and alignment to other remote or eddy-covariance supports remain unresolved. The entire 35-ha target is retained, including the incompletely remeasured central wet-candidate region.

The recorded dated rows span June 2, 2010–March 7, 2014 in E0 and May 17, 2018–January 3, 2020 in E1. A census epoch is a field-observation interval; the reconstruction origin is January 3, 2020. Measured earlier endpoint states are projected over their actual available time intervals, with projection uncertainty. Third-census fieldwork is publicly reported for May 2024–August 2025; the present analysis uses publicly exposed aggregate descriptions in a scoped external consistency comparison [8, 19].

2.2 Identity and observation-status rules

Within the retrieved E0/E1 tables, `stem.id` is a stem identifier and `tree.id` a plant association. Thirty-one shared stems change recorded plant association, and 4,514 shared-stem transitions change taxon code. The reconstruction retains alternative association branches; the unchanged-taxon training fit excludes the ambiguous taxon transitions. Four E0 stem identifiers missing from E1 remain of unknown fate. Dated `alive` and `stem dead` endpoints are used under an explicitly conditional crosswalk. Secure death for the masking study additionally requires an explicit D code and excludes ambiguous Miss/MT/S codes; observed alive requires a corroborating A status. A missing response is carried as latent evidence, with its observational label preserved throughout.

2.3 Evidence and transformation provenance

Each field and calculated output has a source identifier, transformation method, and one or more provenance classes: **OBSERVED** for archived measurements, **DERIVED** for transformations of observed variables, **RECOVERED** for cross-epoch recovered coordinates under stable identity, **IMPUTED** for stochastic latent state, **SIMULATED** for sampled futures, **LITERATURE_LICENSED** for qualitative ecological motivation, and **EXTERNALLY_CONSTRAINED** for demonstration choices. Each provenance class attaches to the contributing fields and transformations of a reconstructed state. In particular, every origin-state cohort count/diameter is inferred at January 2020; the measured-source fraction records observation ancestry; January 2020 origin values remain reconstructed latent quantities.

The 35-ha plot is the common target. The EMS AmeriFlux tower, NEON HARV data, and external disturbance experiments are separately inventoried and support-classified in the empirical data inventory; none enters the demographic fit through an unverified plot-footprint alignment. Later adult-tree outcomes from HF453 are excluded from fitting and are used exclusively for a support-matched independent mortality evaluation. The HF355 seedling series supports separate descriptive analysis of small-plot transitions.

2.4 Independent outcome carriers

The adult component test uses the published HF453 annual tree health and mortality survey, 2021–2024, within the same 35-ha ForestGEO setting [6]. Its observed units are tagged adult stems at least 10 cm DBH and its recorded endpoints distinguish alive, alive-with-issue, secure dead, and dead/not-found categories. The separately sampled HF355 survey follows seedlings below 1 cm DBH in small subplots and records observed survival and field-specific recruitment/graduation indicators [7]. The 1–10 cm whole-plot stem-support variable of the prospective specification is a different carrier. Every external test below specifies a linkage rule, risk set, time window, status crosswalk, and exclusions before comparing outcomes.

3 Formal empirical-synthetic specification

3.1 Objects and semantic separation

Write $Y = (Y_{E0}, Y_{E1})$ for the observed tagged-stem record, X for a latent 35-ha cohort state, d for a declared scenario, and t for a physical horizon measured in years. A candidate history $\gamma = (X_0, \dots, X_t)$ contains annual state transitions produced by a stochastic kernel $G_{\vartheta, d}$. Define

$$V_{d,t}(X_0) = \{\gamma \in \text{Hist}(d, t, X_0) : \text{Lawful}_G(\gamma) \wedge \text{Continues}_{\mathcal{S}}(\gamma)\}, \quad (2)$$

$$C_{\mathcal{S}}(X_0; d, t) = \text{collect}(\{(o(\gamma), w_\gamma) : \gamma \in V_{d,t}(X_0)\}). \quad (3)$$

Here Lawful_G is the implemented *model-admissibility realization* of the formal lawfulness interface: chronological accounting, cohort identity, count bounds, and size bounds. It is an executable integrity predicate for the stated generator. $\text{Continues}_{\mathcal{S}}$ is a separately stipulated structural-demographic path condition. The capacity observer o consumes responses of viable histories and retains trajectory identity and original predictive weight. The raw census observation mechanism $Y \mid X$ and the capacity observer o are different maps.

Empirical state, scenario, and parameter domains. The empirical state space at the reconstructed origin is the measurable set

$$\mathcal{X}_{64} = (\mathbb{N}_0 \times [1, 300])^{64}, \quad \mathcal{D}_{\text{scen}} \subset \mathbb{R}_+^3 \times \mathbb{R}_+, \quad \mathcal{D}_{\text{label}} = \{0, 1, 2\}_p \times \{0, 1, 2\}_\ell. \quad (4)$$

The scenario coordinates $d_{\text{scen}} = (f_h, f_g, f_r, \delta_h)$ modify a conditional generator; the label coordinates (p, ℓ) select hazard and entry parameter-generating laws in controlled synthetic experiments. These are the empirically implemented context coordinates:

$$\mathcal{D}_{\text{emp}} = \mathcal{D}_{\text{scen}} \times \mathcal{D}_{\text{label}}, \quad \mathcal{T}_{\text{eval}} = \{1, 5, 10, 20\} \text{ years}, \quad (5)$$

$$\mathcal{I}_{\text{disclosure}} = \{\text{full, coarsened, unknown, label-only}\}. \quad (6)$$

The evaluation horizon $t \in \mathcal{T}_{\text{eval}}$ and estimator information policy $\iota \in \mathcal{I}_{\text{disclosure}}$ are *separately indexed* from the physical context. The empirical-synthetic experiments use the relevant projections of $\mathcal{D}_{\text{emp}}$: the original scenario run fixes d_{scen} , while the controlled context-label studies also specify (p, ℓ) and disclose only the information allowed by ι . This is a limited implementation of the broader formal context object in Paper II. Table 2 declares four fixed scenario modifiers. Their values are experimental specifications with no fitted climatic sampling distribution. The model parameter space is

$$\Theta = (\mathbb{R}_+ \times [0, .7] \times \mathbb{R}_+ \times \mathbb{R}_+)^{64}, \quad \vartheta = ((h_c, g_c, \sigma_c, \Lambda_c))_{c=0}^{63}. \quad (7)$$

Here g_c is a sampled nominal growth rate, σ_c the fitted increment spread, and Λ_c a sampled candidate-entry intensity. A separate index $m \in \{\Gamma, N^+\}$ selects the future growth kernel. This notation makes the dependence in $G_{\vartheta, d, m}$ explicit.

The demonstration Health proposition is

$$H_{\mathcal{S}, r}(X_0; d, t) = \text{Realizes}_{\mathcal{S}}(X_0) \wedge \text{Adequate}_r(C_{\mathcal{S}}(X_0; d, t)). \quad (8)$$

A declared semantic specification $\mathcal{S}$ records the definition of the carrier, *Continues*, the observer, the response collector, and the adequacy queries. Alternative specifications index distinct questions. The semantic and scenario index sets have no assigned mixing probabilities; each specified case is evaluated conditionally.

3.2 Finite-design measure and predictive measure

The full finite-design analysis uses 256 reconstructed states, two parameter draws per state, and two growth kernels. The resulting 1,024 outer units u have equal design weights $a_u = 1/1024$. Within outer unit u and scenario d , the kernel produces $K = 64$ histories with conditional empirical weights $b_{j|u, d} = 1/K$. Thus each joint history in a fixed scenario has

$$w_{u, j|d} = a_u b_{j|u, d} = (1024K)^{-1}, \quad \sum_{u=1}^{1024} \sum_{j=1}^K w_{u, j|d} = 1. \quad (9)$$

The inner distribution approximates stochastic demographic futures given a reconstructed state and fitted-parameter draw.

Outer and inner measures. Index the state draws by $i = 1, \dots, 256$, the per-state parameter draws by $p \in \{1, 2\}$, and the variants by $m \in \{\Gamma, N^+\}$. The joint discrete outer measure is

$$\nu_Y^{(1024)} = \frac{1}{1024} \sum_{i=1}^{256} \sum_{p=1}^2 \sum_{m \in \{\Gamma, N^+\}} \delta_{(X_i, \vartheta_{ip}^{\text{pred}}, m)}. \quad (10)$$

For a fixed declared scenario d , write $P_{\vartheta, d, m}^{(t)}(d\gamma \mid X)$ for the stochastic history law and $P_{u, d}^{(K)} = K^{-1} \sum_{j=1}^K \delta_{\gamma_{u, d, j}}$ for its sampled finite approximation. Thus

$$\hat{\mathcal{H}}_{\mathcal{S}, r}(d, t) = \int H_{\mathcal{S}, r}^{(K)}(X; d, t, \vartheta, m) \nu_Y^{(1024)}(dX, d\vartheta, dm), \quad (11)$$

$$= \frac{1}{1024} \sum_{u=1}^{1024} \mathbf{1} \left\{ \text{Realizes}_{\mathcal{S}}(X_u) \wedge \text{Adequate}_r(C_{\mathcal{S}}^{(K)}(u, d, t)) \right\}. \quad (12)$$

The superscript (K) specifies a finite bank. In general, $\int \mathbf{1}\{m_V(u) \geq \theta\} d\nu \neq \mathbf{1}\{\int m_V(u) d\nu \geq \theta\}$; the reported finite-design Health fraction uses the former. The outer averaging is an equally weighted finite *design mixture* of reconstructed states, parameter samples and two model variants. The resulting fraction is conditional on those mixing choices. Scenario and semantic priors and a uniquely identified Bayesian posterior remain outside this finite-design construction.

For each outer unit u , fix the *unconditional* empirical candidate law

$$\mu_u^{(K)} = \frac{1}{K} \sum_{j=1}^K \delta_{\gamma_{uj}}$$

on its sampled history bank Γ_u . The collected viable capacity is the *unrenormalized subprobability measure*

$$\mu_{V,u}^{(K)}(A) = \frac{1}{K} \sum_{j=1}^K \mathbf{1}\{\gamma_{uj} \in A\} \mathbf{1}\{\text{Lawful}_G(\gamma_{uj}) \wedge \text{Continues}_S(\gamma_{uj})\}, \quad A \subseteq \Gamma_u. \quad (13)$$

Its total mass, and the mass of its reserve-qualified restriction, are

$$m_V(u, d, t; \mathcal{S}) = \mu_{V,u}^{(K)}(\Gamma_u), \quad (14)$$

$$m_R(u, d, t; \mathcal{S}, \rho) = \int_{\Gamma_u} \mathbf{1}\{R_\rho(\gamma)\} \mu_{V,u}^{(K)}(d\gamma). \quad (15)$$

Failures retain their weights in the *original candidate law*, while a *separate complementary accounting ledger* records

$$m_{\text{unlawful}} = \mu_u^{(K)}(\{\gamma : \neg \text{Lawful}_G(\gamma)\}), \quad (16)$$

$$m_{\text{lawful failure}} = \mu_u^{(K)}(\{\gamma : \text{Lawful}_G(\gamma) \wedge \neg \text{Continues}_S(\gamma)\}), \quad (17)$$

$$1 = m_{\text{unlawful}} + m_{\text{lawful failure}} + m_V. \quad (18)$$

The viable subprobability measure has viable histories as its support. The complementary accounting ledger uses the original measure $\mu_u^{(K)}$ to assign mass to unlawful and lawful noncontinuing histories. All capacity weights retain the original candidate-history denominator. The finite-bank census in (14) is exact for the sampled bank and approximates the specified stochastic-kernel masses.

4 Methods: observation-to-state and demographic laws

4.1 Cohort state and units of analysis

An origin state comprises 64 initial cells indexed by

$$c = 16k + 4s + z, \quad k, s, z \in \{0, 1, 2, 3\}. \quad (19)$$

The taxon groups are hemlock, other canopy species, shrubs, and other taxa; the stratum groups are west, central, east, and the *wet-candidate/protocol proxy*; E0 DBH size classes are $[1, 3)$, $[3, 10)$, $[10, 30)$ and $[30, 300]$ cm. The wet-candidate membership rule selects an E0 living stem in the central local- x third with a November–March field date during 2012–2014; membership remains anchored to E0.

The record-to-cell partition. Let a raw record y_i contain taxon code a_i , local coordinate x_i , exact observation date t_i , diameter D_i , and linked E0 record y_i^0 where available. Define the four taxon sets

$$T_0 = \{\text{tsugca}\}, \quad (20)$$

$$T_1 = \{\text{acerru, querru, pinust, fagugr, betual, betule, betupa, acersa}\}, \quad (21)$$

$$T_2 = \{\text{ilexve, vaccco, kalmla, hamavi, ilexmu, nemomu, lyonli, alnuin, vibuca, vibunu, vibual, vibula}\}, \quad (22)$$

$$T_3 = \mathcal{T}_{\text{raw}} \setminus (T_0 \cup T_1 \cup T_2), \quad k(y_i) = \begin{cases} 0, & a_i \in T_0, \\ 1, & a_i \in T_1, \\ 2, & a_i \in T_2, \\ 3, & \text{otherwise.} \end{cases} \quad (23)$$

Here $\mathcal{T}_{\text{raw}}$ includes the raw code domain and its absent/unknown placeholders. Thus the fallback $k = 3$ is the *computational catch-all* for other, unrecognized, or missing taxon codes; this catch-all assignment carries an explicit unknown/other computational status and no biological taxon identification. The transition fit independently requires an unchanged recorded taxon code; missing and changed codes remain outside the matched-species training subset. The operational coordinate selector and size selector are

$$(x_i^*, t_i^*) = \begin{cases} (x_i, t_i), & x_i \text{ recorded,} \\ (x_i^0, t_i^0), & x_i \text{ absent and E0 link has } x_i^0, \\ (\perp, \perp), & \text{otherwise,} \end{cases} \quad (24)$$

$$D_i^* = \begin{cases} D_i, & D_i \text{ recorded,} \\ D_i^0, & D_i \text{ absent and E0 DBH recorded,} \\ 1.5, & \text{otherwise,} \end{cases} \quad (25)$$

$$z_i = \mathbf{1}\{D_i^* \geq 3\} + \mathbf{1}\{D_i^* \geq 10\} + \mathbf{1}\{D_i^* \geq 30\}. \quad (26)$$

Let $w(y) = \mathbf{1}\{x^* \in [700/3, 1400/3), \text{year}(t^*) \in \{2012, 2013, 2014\}, \text{month}(t^*) \in \{11, 12, 1, 2, 3\}\}$, with $w = 0$ whenever the selected coordinate or relevant date is missing or does not have the recognized year/month fields. This is a recorded-date string-field classification with month information supplied by the selected recorded source. In particular, when an E1 coordinate is recorded with a missing date, new E1 proxy membership is unestablished and existing E0 proxy membership is retained through the first branch of Eq. (27). The implementation's stratum map is

$$s(y_i; y_i^0) = \begin{cases} 3, & y_i^0 \text{ exists and } s(y_i^0; \emptyset) = 3, \\ 3, & w(y_i) = 1, \\ 1, & x_i^* = \perp, \\ \min\{2, \max\{0, \lfloor 3x_i^*/700 \rfloor\}\}, & \text{otherwise.} \end{cases} \quad (27)$$

The resulting measurable deterministic cell map is

$$\pi(y_i; y_i^0) = (k(y_i), s(y_i; y_i^0), z_i), \quad c(y_i; y_i^0) = 16k(y_i) + 4s(y_i; y_i^0) + z_i. \quad (28)$$

Class boundaries 3, 10, 30 are assigned to the upper class. The E0 living cohort uses its E0 record; E1 status/measurement aggregation uses its E1 record when available, with the stated E0 coordinate/DBH fallbacks. The date accompanying x_i^* is chosen from the same source record

as the coordinate; new wet-proxy membership requires recognized year and month fields in the selected record. Stable E0 proxy membership is carried forward. When a shared stem changes tree association, both recorded cell assignments are retained as alternatives for the association branch; tree/taxon ambiguity remains an explicit alternative association in the reconstructed state. Missing location receives the computational default stratum 1 and retains its missing-coordinate provenance. The literal code sets and fallback rules supply the complete partition map in Eq. (19). It is a sampling- and location-derived proxy whose hydrological status remains uncertified. A stem belongs to one model stratum, and the entire 35-ha population remains in the model.

State $X = (n_c, d_c)_{c=0}^{63}$ stores integer cohort counts and a root-mean-square DBH. The index convention $c \in \{0, \dots, 63\}$ is used throughout and matches the 64-cell computational arrays. Its total basal area and juvenile support are

$$\text{BA}(X) = \frac{\pi}{40000} \sum_{c=0}^{63} n_c d_c^2 \quad (\text{m}^2), \quad J(X) = \sum_{c=0}^{63} n_c \mathbf{1}\{d_c < 10 \text{ cm}\}. \quad (29)$$

The conversion follows diameter in centimeters and circular basal area. J is a 1–10 cm size-class proxy for young woody stems and records juvenile woody-stem support. Seed production, seedling establishment, and fecundity require their own measurement and licensing. An RMS cohort diameter retains the second moment contributing to basal area, with individual-size distribution and threshold-crossing detail compressed. Birth cohorts are separately tracked by time and taxon/stratum during future simulation.

4.2 Transition eligibility, censoring, and chronology

The training reader pairs recorded stems by `stem.id`, checks stable `tree.id` and unchanged taxon code, uses actual positive inter-census duration Δ_i in years, and assigns secure interpreted status where available. Unresolved status is retained for reconstruction. For an alive paired stem, a diameter increment enters growth fitting when measurement-height difference is at most 0.05 m after the convention $0 \mapsto 1.3$ m, and the annualized recorded increment lies between -0.1 and 1 cm/year. These selection rules produce 69,158 interpreted survival endpoints and 53,361 growth increments in the declared fit. The 240 growth exclusions are quarantined as a defined fitting exclusion, and taxon-code changes remain visible in the QA ledger.

Fitted survival data represent secure endpoint status of the same tagged stem. They are distinct from whole-plant disappearance or categorical mortality of the 35-ha organization. The stricter masked-data evaluation pool imposes additional corroboration and contains 66,284 eligible stem transitions. These different counts refer to different documented selection filters.

4.3 Hazard identification from unequal intervals

For cell c , let $Z_i = 1$ denote an interpreted surviving stem after Δ_i years and use an annual constant hazard h_c . Its endpoint survival probability is

$$P(Z_i = 1 \mid h_c, \Delta_i) = e^{-h_c \Delta_i}. \quad (30)$$

The demonstrative pooled annual hazard is $\bar{h} = -\log(\hat{p}_{\text{pool}})/\bar{\Delta}$, where $\hat{p}_{\text{pool}}$ is the pooled endpoint survival fraction. The fitted hazard grid has 256 equally spaced values in

$$\mathcal{H} = [-\log(0.9995), -\log(0.5)]. \quad (31)$$

With annual survival $q_c = e^{-h_c}$, the prior is beta-shaped in q_c with strength 32 and shape parameters $a = 32e^{-\bar{h}}$ and $b = 32 - a$. The hazard-grid log prior, including its transformation Jacobian, is

$$\ell_{0,c}(h) = -ah + (b - 1) \log(1 - e^{-h}). \quad (32)$$

The observed unequal-interval endpoint log likelihood is

$$\ell_c(h) = \sum_{i:Z_i=1} (-h\Delta_i) + \sum_{i:Z_i=0} \log(1 - e^{-h\Delta_i}). \quad (33)$$

To expose dependence and limit pseudoreplication, the demonstration tempers this likelihood by $\kappa_c = \min\{1, 200/n_c^{\text{fit}}\}$. The sampled hazard grid is

$$\hat{\pi}_c(h_g) \propto \exp\{\ell_{0,c}(h_g) + \kappa_c \ell_c(h_g)\}. \quad (34)$$

The cap of 200 is a declared methodological assumption. Identification of the effective sample size under spatial stem dependence remains a separate statistical task. The recorded grid/probability array is saved for every cell. Cell survival in the fitted law is conditional on this model and the source-status crosswalk.

4.4 Growth and candidate threshold entry

Let $g_{ci} = (D_{E1,i} - D_{E0,i})/\Delta_i$ for accepted growth intervals, with $\bar{g}$ the pooled accepted annual growth mean. The cell mean, standard deviation and standard error are defined computationally as

$$\hat{\mu}_c = \text{clip}_{[0,0.7]} \left(\frac{\sum_i g_{ci} + 32\bar{g}}{n_c^g + 32} \right), \quad (35)$$

$$\hat{\sigma}_c = \max\{0.02, \text{sd}(g_c)\}, \quad \hat{\text{se}}_c = \frac{\hat{\sigma}_c}{\sqrt{\min\{\max(n_c^g, 1), 200\}}}. \quad (36)$$

Sparse cells use the documented pooled spread fallback. Parameter sampling clips a normal draw centered at $\hat{\mu}_c$ into $[0, 0.7]$. Two future growth kernels then use either nonnegative gamma increments with nominal μ_c and σ_c , or a reflected/clipped normal increment $\max\{0, N(\mu_c, \sigma_c^2)\}$. Their effective mean growth can differ after reflection.

The E1-only living stem count in each cell supplies e_c , a *candidate* threshold-entry quantity. The nominal six-year exposure gives the recorded-candidate rate $r_c^{\text{obs}} = e_c/6$. In the unremeasured wet-candidate/protocol stratum, the candidate-entry model pools *all size classes* of recorded candidates in the three nonproxy strata for each taxon group k , then scales that pooled rate by the E0 proxy-to-nonproxy stem-count ratio:

$$r_{k,3,0}^{\text{model}} = \left(\sum_{s=0}^2 \sum_{z=0}^3 r_{k,s,z}^{\text{obs}} \right) \frac{\sum_{z=0}^3 n_{k,3,z}^0}{\max\{1, \sum_{s=0}^2 \sum_{z=0}^3 n_{k,s,z}^0\}}, \quad (37)$$

$$r_c^{\text{model}} = r_c^{\text{obs}} \quad \text{for all other cells}; \quad r_c := r_c^{\text{model}} \quad \text{below.}$$

The entire borrowed rate is assigned to the *single* proxy cell $z = 0$; the proxy-specific size-class recruitment shares remain unidentified. Other proxy cells retain their recorded candidate rates (zero for this unremeasured support). This assumption extrapolates a count ratio across protocol-defined strata. The actual wet-region entry process and its ecological interpretation remain empirically

unidentified. A positive pseudo-exposure is retained even when the recorded candidate count is zero. For a parameter draw, the entry intensity is sampled from

$$\Lambda_c \sim \text{Gamma}\left(k_c, \frac{r_c + 1/6}{k_c}\right), \quad k_c = \min\{6r_c + 1, 200\}, \quad (38)$$

using shape–scale convention. This is a demonstrative uncertainty model conditional on an unresolved entry-identification mechanism. The identification experiment in Sec. 5.3 tests the evidence available for that mechanism.

Entry observation and modeled birth. The observation variable $e_{k,s,z}$ counts stems first recorded in E1 and assigned to size class z from their *recorded E1 diameter*, subject to the study’s source-coordinate and identity rules. Their unobserved E0 diameters are not reconstructed as direct measurements for this assignment. The future generator instead draws annual cohorts indexed by (k, s) with initial simulated diameter 1.7 cm. Its entry intensity aggregates the fitted size-class intensities,

$$\lambda_{k,s}^{\text{birth}} = \sum_{z=0}^3 \Lambda_{16k+4s+z}, \quad B_{k,s,t+1} \mid \vartheta, d \sim \text{Poisson}(f_r^{(d)} \lambda_{k,s}^{\text{birth}}), \quad D^{\text{birth}} = 1.7 \text{ cm}. \quad (39)$$

Thus observed first appearance and simulated birth at 1.7 cm are linked by a stipulated observation-to-process rule. Their biological event equivalence is unidentified. For a size-restricted candidate-input diagnostic, define $e_{k,s,z}^{(<3)} = e_{k,s,z} \mathbf{1}\{z = 0\}$ and refit the candidate-intensity law using the same rule. That definition is a counterfactual input, requiring a separately executed prospective propagation before any changed Health outcomes can be reported. The observed class counts and the magnitude of this raw-input restriction are reported in Sec. 6.3.

A single fitted sampling law. Write $F(Y)$ for the deterministic 64-cell sufficient statistics, including the pooled rates, hazard-grid probabilities $\hat{\pi}_c$, empirical growth summaries, and candidate-entry counts. For a given cap q , the fitted sampler defines

$$\begin{aligned} \Pi_{Y,q}(d\vartheta) = \prod_{c=0}^{63} \bigg[& \left(\sum_{v=1}^{256} \hat{\pi}_{c,q}(h_v) \delta_{h_v}(dh_c) \right) \mathcal{L}\left(\text{clip}_{[0,7]}[Z_c]\right) (dg_c) \\ & \times \delta_{\hat{\sigma}_c}^{\sim}(d\sigma_c) \text{Gamma}\left(k_{c,q}, \frac{r_c + 1/6}{k_{c,q}}\right) (d\Lambda_c) \bigg], \end{aligned} \quad (40)$$

where $Z_c \sim N(\hat{\mu}_c, \hat{\text{se}}_{c,q}^2)$ and $k_{c,q} = \min\{6r_c + 1, q\}$. The distribution is conditional on the single pooled fit $F(Y)$, which couples the estimated cell summaries through the shared source data; random draws of individual cell components are then independent given that fit. The numerical distribution is the declared finite-grid/Gamma/normal sampling law, conditional on the pooled fit. Its Bayesian joint-posterior interpretation remains unidentified. The principal fit fixes $q = 200$; sensitivity analyses evaluate $q \in \{50, 100, 200, 500\}$. For each reconstructed X_i , two independent *predictive* parameter draws $\vartheta_{ip}^{\text{pred}} \sim \Pi_{Y,200}$ are taken and both growth variants applied to each draw. A distinct *reconstruction* parameter $\vartheta_i^{\text{rec}} \sim \Pi_{Y,200}$ is already sampled inside the generation of X_i . The two stages use the same fitted sampling measure with independent realized parameter draws conditional on $F(Y)$.

4.5 Individual-exposure reconstruction kernel

Let $T_0 = 2020-01-03$ be the modeled origin, and index the recorded stem contributions in each cell c by $i \in \mathcal{I}_c$. The record carries an observed-source class $a_i \in \{\text{measured, unknown, alive-missing-size}\}$ and an actual recorded elapsed time Δ_i from its applicable census observation to T_0 . Measured living stems are projected from their later census date; unknown-fate stems retain their earlier recorded date; living stems with missing DBH retain observed living status and an imputed diameter. Each contribution's survival indicator is

$$\begin{aligned} A_i \mid h_c, \eta, \eta_w, Y &\sim \text{Bernoulli}(p_i), \\ p_i &= \exp\left[-h_c \exp\left\{\mathbf{1}_{\{a_i=\text{unknown}\}}(\eta + I_w(c)\eta_w)\right\} \Delta_i\right], \\ \eta &\sim N(0, 0.45^2), \quad \eta_w \sim N(0, 0.7^2). \end{aligned} \quad (41)$$

The shared random effects apply to unresolved-fate components and induce conditional dependence after marginalization; the Bernoulli factors are conditionally independent given hazards and the shared shifts. This is the *individual-time reconstruction* used for the main corrected analysis. Unknown fate remains an observed unresolved-status category throughout the reconstruction.

For each source-status component a within cohort c , condition on the sampled cell hazard and the shared missingness shifts $\zeta = (\eta, \eta_w)$. Its effective hazard is constant within that component,

$$\lambda_{a,c}(h_c, \zeta) = h_c \exp\left[\mathbf{1}_{\{a=\text{unknown}\}}\{\eta + I_w(c)\eta_w\}\right], \quad p_i = e^{-\lambda_{a,c}\Delta_i} \quad (i \in \mathcal{I}_{a,c}). \quad (42)$$

The source-status-matched pooled-time counterfactual retains the same $\lambda_{a,c}$ and component membership but replaces the actual exposures by their component mean:

$$N_{a,c}^{\text{pool}} \mid h_c, \zeta \sim \text{Binomial}\left(n_{a,c}, e^{-\lambda_{a,c}\bar{\Delta}_{a,c}}\right), \quad \bar{\Delta}_{a,c} = \frac{1}{n_{a,c}} \sum_{i \in \mathcal{I}_{a,c}} \Delta_i. \quad (43)$$

The individual-time conditional survivor count is a Poisson–binomial sum; for $N_c^{\text{ind}} = \sum_a \sum_{i \in \mathcal{I}_{a,c}} A_i$,

$$\begin{aligned} \mathbb{E}[N_c^{\text{ind}} \mid h_c, \zeta, Y] &= \sum_a \sum_{i \in \mathcal{I}_{a,c}} p_i, \\ \text{Var}(N_c^{\text{ind}} \mid h_c, \zeta, Y) &= \sum_a \sum_{i \in \mathcal{I}_{a,c}} p_i(1 - p_i). \end{aligned} \quad (44)$$

Its marginal law over ζ is a mixture of these conditional laws. Writing $M_c(h_c, \zeta) = \sum_i p_i$ and $V_c(h_c, \zeta) = \sum_i p_i(1 - p_i)$, the law of total variance gives

$$\begin{aligned} \mathbb{E}[N_c^{\text{ind}} \mid h_c, Y] &= \mathbb{E}_\zeta[M_c(h_c, \zeta)], \\ \text{Var}(N_c^{\text{ind}} \mid h_c, Y) &= \mathbb{E}_\zeta[V_c(h_c, \zeta)] + \text{Var}_\zeta[M_c(h_c, \zeta)]. \end{aligned} \quad (45)$$

For each fixed h_c, ζ and component with common effective hazard, Jensen's inequality yields

$$\sum_{i \in \mathcal{I}_{a,c}} e^{-\lambda_{a,c}\Delta_i} \geq n_{a,c} e^{-\lambda_{a,c}\bar{\Delta}_{a,c}}. \quad (46)$$

Summing the component-level inequality preserves the conditional mean ordering for the matched effective-hazard construction. The inequality concerns expected counts; individual sampled counts

are unordered. Integrating over shared shifts requires the expectations in Eq. (45). The exact Poisson–binomial numerical fixtures condition on the sampled h_c, η, η_w and exposure records; they do not posit unconditional stem independence.

Component diameters use the recorded census RMS anchor $\bar{d}_{a,c}$, a sampled cell growth rate g_c , and a component-specific normal disturbance. For the observed measured-source component $\sigma_a = 0.1$ cm, while unresolved and alive-without-size components use $\sigma_a = 1$ cm. The implementation samples a single cell/component disturbance and applies each surviving record’s actual exposure:

$$D_i^* = \text{clip}_{[1,300]}(\bar{d}_{a_i,c} + g_c \Delta_i + \varepsilon_{a_i,c}), \quad \varepsilon_{a,c} \sim N(0, \sigma_a^2). \quad (47)$$

The common component perturbation is part of the stated model: it is not an independent stem-level size measurement. For the wet-candidate/protocol stratum, supplemental unobserved counts follow

$$U_c \sim \text{Poisson}(6\Lambda_c I_w(c)e^\epsilon), \quad \epsilon \sim N(0, 0.35^2), \quad D_c^{\text{new}} = 1.7 \text{ cm}. \quad (48)$$

Thirty-one ambiguous plant-association cases each receive one of the recorded alternative assignments under an explicit equal-branch draw; no record is duplicated. Recorded E1 diameter and a known corroborated fate are used on the corresponding branch where available. The same original observation and identity rules are retained in both exposure variants.

Let $\mathcal{A}_c$ enumerate measured, unresolved, alive-missing-size, wet entry and association contributions. With count $N_{c,a}$ and per-contribution squared-diameter total $S_{c,a}$, the reconstructed cohort state is the deterministic map

$$n_c = \sum_{a \in \mathcal{A}_c} N_{c,a}, \quad d_c^2 = \begin{cases} (\sum_a S_{c,a})/n_c, & n_c > 0, \\ \bar{d}_{0,c}^2, & n_c = 0. \end{cases} \quad (49)$$

It retains the reconstructed count and basal-area second moment. It does not retain every individual diameter threshold classification. All origin-state values remain inferred even where their source ancestry is directly observed.

Unified sampling kernel. Writing Z for the sampled shared shifts, Bernoulli/Poisson counts, size components and association assignments, and F_{RMS} for Eq. (49), the reconstruction law is the explicit pushforward

$$K_{\text{rec}}(A \mid Y, \vartheta) = \int \mathbf{1}\{F_{\text{RMS}}(Z) \in A\} Q_{Y,\vartheta}(dZ), \quad (50)$$

Here $Q_{Y,\vartheta}$ is the conditional product/shared-latent law specified above. The fitted observed-data-to-state mixture is

$$K_{Y,q}(A) = \int K_{\text{rec}}(A \mid Y, \vartheta) \Pi_{Y,q}(d\vartheta). \quad (51)$$

Reconstruction and future parameters have the same fitted marginal law and are sampled independently conditional on its summary statistics. The study propagates this explicit model law, whose biological identification and calibration are evaluated in Results.

4.6 Annual demographic trajectory generator

Each scenario and outer unit generates K trajectories over 20 annual steps. At the beginning of year t , each existing cohort has count $n_{c,t}$ and RMS diameter $d_{c,t}$. For taxon group $k(c)$, the scenario annual hazard is

$$h_c^{(d)} = f_h^{(d)} h_c + I\{k(c) = \text{hemlock}\} \delta_h^{(d)}. \quad (52)$$

Then

$$S_{c,t+1} \sim \text{Bin}(n_{c,t}, e^{-h_c^{(d)}}), \quad (53)$$

$$D_{c,t+1} = n_{c,t} - S_{c,t+1}, \quad (54)$$

$$d_{c,t+1} = \text{clip}_{[1,300]}(d_{c,t} + G_{c,t}^{(d)}), \quad (55)$$

$$B_{k,s,t+1} \sim \text{Poisson}(f_r^{(d)} \lambda_{k,s}^{\text{birth}}). \quad (56)$$

$G_{c,t}^{(d)}$ is a gamma or nonnegative-normal growth increment scaled by $f_g^{(d)}$; each cohort receives one shared growth increment, as specified by the cohort-resolution generator. Sixteen fresh (k, s) birth cohorts are created annually, with initial 1.7-cm diameter and distinct birth-time identities. Existing and new cohorts remain separately indexed. Deaths and threshold entries accumulate, and annual population accounting is exactly

$$N_{t+1} = N_t - D_{t+1} + B_{t+1}. \quad (57)$$

The outcome ledger records total count, basal area, J_t , hemlock basal area, cumulative cohort deaths, and cumulative entries at every annual checkpoint.

The exact two-variant increment law. For the parameter draw g_c, σ_c and scenario growth factor f_g , set $\mu'_c = f_g g_c$, $s'_c = f_g \sigma_c$. At each annual checkpoint and for each extant cohort identity a with parent cell $c(a)$, the specified increment law is

$$G_{a,t}^{(\Gamma,d)} \sim \begin{cases} \text{Gamma}(\alpha_c, \beta_c), & \mu'_c > 0, s'_c > 0, \\ \delta_{\mu'_c}, & s'_c = 0, \\ \delta_0, & \mu'_c = 0, \end{cases} \quad (58)$$

$$\alpha_c = \max\{(\mu'_c/s'_c)^2, 10^{-6}\}, \quad \beta_c = \mu'_c/\alpha_c \quad (\mu'_c > 0, s'_c > 0), \quad (59)$$

$$G_{a,t}^{(N^+,d)} = \max\{0, Z_{a,t}\}, \quad Z_{a,t} \sim N(\mu'_c, (s'_c)^2). \quad (60)$$

The positive Gamma law is in shape–scale form. It satisfies $\mathbb{E}G = \mu'_c$ and $\text{Var}(G) = (\mu'_c)^2/\alpha_c$; its variance equals $(s'_c)^2$ off the shape floor. Nonnegative growth with zero mean must be degenerate at zero; the $s'_c = 0$ case is degenerate at its positive mean. The regularization preserves the specified mean and changes the variance when the shape floor is active. The original shape-floored alternative with scale $(s'_c)^2/\max(\mu'_c, 10^{-9})$ is included solely in the registered law-sensitivity comparison. Its moments are calculated from its actual shape and scale.

The Gamma is in shape–scale form. The second law is a rectified normal, whose negative pre-rectification values contribute an atom at zero. The scenario factor rescales μ'_c and s'_c before drawing the increment, and $d_{a,t+1} = \text{clip}_{[1,300]}(d_{a,t} + G_{a,t})$. The nonnegative-normal effective mean is

$$\mathbb{E}[G^{(N^+,d)}] = s'_c \phi(\mu'_c/s'_c) + \mu'_c \Phi(\mu'_c/s'_c) \quad (s'_c > 0), \quad (61)$$

Scenario	interpretation	f_h	f_g	f_r	δ_h
D0	Reference synthetic forcing	1.00	1.00	1.00	0
D1	Drought-like synthetic stress	1.60	0.65	0.65	0
D2	Hemlock-pressure synthetic stress	1.00	0.85	0.90	0.08
D3	Combined synthetic stress	1.80	0.50	0.50	0.08

Table 2: Complete scenario multipliers for the stated cohort generator. General hazard factors multiply fitted hazards; δ_h is an *additive annual* hemlock excess hazard. All values are explicit demonstration assumptions, with ecological mechanisms motivated by the disturbance literature [13, 14, 15].

which differs from μ'_c in general; diameter clipping introduces further boundary effects for either variant. One growth increment is drawn per cohort identity, so individual stems within that cohort share the sampled increment. This is the exact distinction between the implemented variants.

Model-admissibility predicate. Let the simulated history ledger at annual checkpoints be $\gamma = (n_{a,t}, d_{a,t}, J_t, N_t, B_t, D_t, C_t^B, C_t^D)_{a,t}$ with separately tracked unique cohort identities a . Define

$$A_1(\gamma) : \quad t = 0, 1, \dots, T \text{ in increasing annual order,} \quad (62)$$

$$A_2(\gamma) : \quad n_{a,t} \in \mathbb{N}_0, \quad 1 \leq d_{a,t} \leq 300, \quad n_{a,t}, d_{a,t} \text{ finite,} \quad (63)$$

$$A_3(\gamma) : \quad a \neq a' \text{ for distinct extant cohort slots;} \quad (64)$$

$$A_4(\gamma) : \quad N_t = N_{t-1} - (C_t^D - C_{t-1}^D) + (C_t^B - C_{t-1}^B), \quad (65)$$

$$A_5(\gamma) : \quad C_t^B \geq C_{t-1}^B, \quad C_t^D \geq C_{t-1}^D, \quad (66)$$

$$A_6(\gamma) : \quad 0 \leq J_t \leq N_t, \quad (N_t, BA_t, J_t, C_t^B, C_t^D) \text{ finite/nonnegative,} \quad (67)$$

$$\text{Lawful}_G(\gamma) \iff \bigwedge_{j=1}^6 A_j(\gamma). \quad (68)$$

The software also validates cohort-specific count/diameter arrays before aggregation, and the ledger includes hemlock basal area as a finite nonnegative field. The cumulative quantities in A_4 count all births and deaths through time, with zero initial cumulative totals. Lawful_G certifies internal admissibility of the generated histories. Statistical validation of the ecological transition law and empirical agreement with forest succession are separate obligations.

The model-admissibility checker tests positive annual chronology, integer nonnegative counts, cohort identity uniqueness, diameter bounds, nondecreasing cumulative deaths and entries, and Eq. (57). The implemented generator is evaluated against these invariants on each candidate path. The checker instantiates the formal **Lawful** interface at the level of this software model; ecological adequacy of the dynamical equations is evaluated separately.

End-to-end conditional sampling law. The empirical-synthetic model is represented by one probability kernel. Let $G_{\vartheta,d,m}^{[t]}$ denote the law of the full t -year path generated by iterating the annual count/diameter/entry kernel $G_{\vartheta,d,m}^{(1)}$, including birth-cohort identities:

$$G_{\vartheta,d,m}^{[t]}(d\gamma \mid X_0) = \delta_{X_0}(dX_0) \prod_{u=0}^{t-1} G_{\vartheta,d,m}^{(1)}(dX_{u+1} \mid X_u). \quad (69)$$

For an E0/E1 evidence object Y , the idealized sampling design corresponding to the declared finite-design study is

$$\begin{aligned} \mathbb{P}_Y^{(d,t)}(d\vartheta^{\text{rec}}, dX, d\vartheta^{\text{pred}}, dm, d\gamma) &= \Pi_{Y,200}(d\vartheta^{\text{rec}}) K_{\text{rec}}(dX \mid Y, \vartheta^{\text{rec}}) \\ &\quad \times \Pi_{Y,200}(d\vartheta^{\text{pred}}) \left[\frac{1}{2}\delta_{\Gamma} + \frac{1}{2}\delta_{N+} \right](dm) \\ &\quad \times G_{\vartheta^{\text{pred}},d,m}^{[t]}(d\gamma \mid X). \end{aligned} \quad (70)$$

The two parameter roles are now separately carried in the joint law: ϑ^{rec} is the parameter draw used for the origin reconstruction, and ϑ^{pred} is the independent predictive draw used for the annual futures. Both have the same fitted marginal measure $\Pi_{Y,200}$ conditional on $F(Y)$; the original four-variable law is their marginal after integrating out ϑ^{rec} . The 256/2/2/64 finite sampling design in Eqs. (10) and (9) is a concrete Monte Carlo realization of these conditional stages, with its imposed equal design weights. Equation (70) is a fully specified *model law*; ecological licensing would additionally require process identification and prospective predictive validation.

4.7 Structural-demographic realization, path continuation, and queries

The declared present-carrier choice is R_1 : relative living basal area and juvenile-cohort support. Let $B_0 = 1478.624132 \text{ m}^2$ and $J_0 = 82077$ be the declared E0 living basal-area and 1–10 cm support denominators. For (α, β) define the annual organization indicator

$$R_{\alpha,\beta}(X_t) = \mathbf{1} \left\{ \frac{\text{BA}(X_t)}{B_0} \geq \alpha \right\} \mathbf{1} \left\{ \frac{J(X_t)}{J_0} \geq \beta \right\}. \quad (71)$$

Let $L_t(\gamma)$ denote the longest *contiguous* run of annual checkpoints up to t at which $R_{\alpha,\beta} = 0$; time zero is included. The declared continuation rule is

$$\text{Continues}_{\alpha,\beta,\tau}(\gamma) \iff R_{\alpha,\beta}(X_t) = 1 \wedge L_t(\gamma) \leq \tau. \quad (72)$$

A trajectory may regain realization before the horizon when the longest contiguous interruption satisfies the declared bound; repeated interruptions are evaluated through their maximum duration. Subannual changes are outside this annual-time model. Taxonomic turnover and stem mortality are permitted under the generic forest carrier when Eq. (72) holds. These parameters declare a demonstration question; ecological acceptance thresholds for the target plot require independent constitutive and observational licensing.

Exact one-year failure decomposition. For an initially realizing state and the special case $(t, \tau) = (1, 0)$, the continuation predicate has a particularly simple normal form:

$$\text{Continues}_{\alpha,\beta,0}(\gamma) \iff [\text{BA}(X_1) \geq \alpha B_0] \wedge [J(X_1) \geq \beta J_0]. \quad (73)$$

Indeed, the initial realization is true and the only remaining checkpoint is the terminal year. Consequently the longest-excursion condition introduces no additional, independent cause of failure for this design. For a bank of K candidate histories, write $\mathcal{L}$ for the model-lawful set, and within $\mathcal{L}$ let F_J and F_B denote terminal juvenile and basal-area threshold failures. The disjoint count partition is

$$\begin{aligned} K &= n_{\neg\mathcal{L}} + n_{\mathcal{L} \cap F_J \cap F_B^c} + n_{\mathcal{L} \cap F_J^c \cap F_B} \\ &\quad + n_{\mathcal{L} \cap F_J \cap F_B} + n_{\mathcal{L} \cap F_J^c \cap F_B^c}, \quad n_V = n_{\mathcal{L} \cap F_J^c \cap F_B^c}. \end{aligned} \quad (74)$$

The corresponding terminal failure indicators are $f_{Jk} = \mathbf{1}\{J(X_{1k}) < \beta J_0\}$ and $f_{Bk} = \mathbf{1}\{BA(X_{1k}) < \alpha B_0\}$. With $\ell_k = \mathbf{1}\{\text{Lawful}_G(\gamma_k)\}$, the four disjoint nonviable counts and the viable count can be recovered directly from the K terminal records:

$$\begin{aligned}
n_{\neg\mathcal{L}} &= \sum_{k=1}^K (1 - \ell_k), \\
n_{J \text{ only}} &= \sum_k \ell_k f_{Jk} (1 - f_{Bk}), \\
n_{B \text{ only}} &= \sum_k \ell_k (1 - f_{Jk}) f_{Bk}, \\
n_{JB} &= \sum_k \ell_k f_{Jk} f_{Bk}, \\
n_V &= \sum_k \ell_k (1 - f_{Jk}) (1 - f_{Bk}).
\end{aligned} \tag{75}$$

Equation (75) is an exhaustive algebraic classification of candidate histories, with unlawful outcomes separated before testing constitutive continuation and simultaneous terminal failures retained in n_{JB} . The five categories can be counted only from pathwise lawfulness flags and terminal juvenile and basal-area records. An independent replay evaluates these indicators directly on every candidate history in the selected one-year design. Both the disjoint partition and the unrestricted, overlapping cause indicators are retained. The classification is checked against the original unconditional candidate counts, stored initial states, and historical viable counts; its numerical result is reported in Sec. 6.4. The classification identifies the terminal predicate responsible for failure in this model and selected bank, with its interpretation conditional on the reconstructed state and declared demographic law.

For a viable history define the terminal juvenile reserve $r_J(\gamma) = J(X_t)/J(X_0)$, taking the implemented value zero when the denominator is zero. With m_V from Eq. (14), the reserve-qualified mass is

$$m_R(\rho) = \sum_j b_j \mathbf{1}\{\text{Lawful}_G(\gamma_j) \wedge \text{Continues}(\gamma_j) \wedge r_J(\gamma_j) \geq \rho\}. \tag{76}$$

The two requirement families are

$$Q_1(\theta) : \quad m_V \geq \theta, \tag{77}$$

$$Q_2(\theta, \rho) : \quad m_R(\rho) \geq \theta. \tag{78}$$

Because $\{\gamma : V(\gamma) \wedge R_\rho(\gamma)\} \subseteq \{\gamma : V(\gamma)\}$, monotonicity of the original nonnegative candidate measure gives $m_R \leq m_V$. Hence $[m_V \geq \theta] \wedge [m_R \geq \theta]$ is exactly equivalent to the single reserve condition in Eq. (77), for arbitrary nonnegative history weights. The adequacy threshold is inclusive. The conditional Health Boolean of one outer unit is the conjunction of initial $R_{\alpha,\beta}(X_0)$ and Eq. (77); the population of outer-unit Boolean answers is reported as an equally weighted finite-design fraction. The unit value at $\theta = 0$ represents a degenerate adequacy boundary, with the separate present-realization requirement retained.

The full finite-design grid is $\alpha \in \{0.5, 0.75, 1\}$, $\beta \in \{0.1, 0.4, 0.8\}$, $\tau \in \{0, 2, 5\}$ years, $\theta \in \{0, 0.5, 0.75, 0.9, 1\}$, $\rho \in \{0.5, 1\}$ for Q2, four scenarios and horizons 1, 5, 10, 20 years. Q1 and the two Q2 variants make three query forms, giving

$$4 \cdot 4 \cdot 3 \cdot 3 \cdot 3 \cdot 5 \cdot 3 = 6480 \tag{79}$$

phase specifications. Their fractions represent a declared design ensemble, indexed separately by scenario and semantics.

5 Methods: empirical tests, synthetic validation, and statistical inference

5.1 Real-data leave-observed-out reconstruction test

The masked-observation study tests reconstruction on source records for which selected E1 outcomes are actually recorded. A secure eligible record requires an E0 living stem, unchanged stem–plant association and taxon, positive E0-to-E1 interval, and corroborated E1 endpoint coding: **alive/A** or **stem dead** with a D code and without Miss, MT or S. These criteria yield 66,284 eligible transitions. The evaluator retains the full observed value; the estimator receives a separately written visible file with all held-out measurement fields physically replaced by NA. Fields removed include status, codes, DBH, coordinates, quadrat, measurement height, AGB and measurement identifiers. Stem, tree, tag, and taxon identifiers remain for the permitted link. E1 campaign month and E0-cell missingness frequency are predeclared assignment covariates; the masking assignments depend exclusively on the permitted covariates, with true E1 fate and DBH held for downstream scoring.

For eligible record i with E0 taxon-group index k_i , size-class index z_i , stratum s_i , campaign month m_i , and observed E0-cell unresolved-fate fraction u_i , the four predeclared masking probabilities are

$$p_{M0,i} = 0.15, \quad (80)$$

$$p_{M1,i} = 0.08 + 0.08k_i + 0.035z_i, \quad (81)$$

$$p_{M2,i} = (0.10, 0.16, 0.25, 0.55)_{s_i}, \quad (82)$$

$$p_{M3,i} = \text{clip}_{[.05,.8]} [(0.10 + 0.65u_i)\{1 + 0.1(m_i \bmod 3 - 1)\}]. \quad (83)$$

Sanitized observation operator. For family $j \in \{0, 1, 2, 3\}$, draw independent internally predeclared $U_i \sim \text{Unif}[0, 1]$ and define $M_{ji} = \mathbf{1}\{U_i < p_{Mj,i}\}$. The exact permissible identity-field set is

$$F_{\text{id}} = \{\text{stem.id}, \text{tree.id}, \text{tag}, \text{stem.tag}, \text{sp}\}.$$

The physical mask is

$$[\mathcal{M}_j(Y; U)]_{i,f} = \begin{cases} Y_{i,f}, & M_{ji} = 0 \text{ or } f \in F_{\text{id}}, \\ \text{NA}, & M_{ji} = 1 \text{ and } f \notin F_{\text{id}}. \end{cases} \quad (84)$$

The predeclared E1 campaign month enters as a masking/design covariate and as an independently supplied known exposure for endpoint scoring; for a held-out row its original **exact.date** is also set to NA in the fitting CSV. The estimator is measurable with respect to $\sigma(\mathcal{M}_j(Y; U), Y_{E0}, \text{allowed exposure covariates}, \xi)$, with fresh RNG ξ independent of the concealed truth given the visible inputs. The evaluator alone retains the hidden endpoint values. This establishes the estimator’s permitted information sigma-algebra and its no-leakage condition. Dependence between genuinely missing outcomes and visible predictors remains an empirical modeling question.

Each mechanism is independently replicated three times with predeclared random streams. After the mask is written, the declared reader, fitting routine, and 256-draw reconstruction are rerun using the

sanitized data. The projected origin reconstruction is retained as the model’s inference product. To score actual 2018–2020 E1 observations, the same sampled unknown-fate and cohort-RMS diameter kernel is evaluated at each stem’s actual E1 exposure interval; observed 2018 or 2019 truth is never mislabeled as the 2020 origin state.

For held-out fate $z_i \in \{0, 1\}$ and predicted survival probability p_i , the per-record Brier score and log loss are

$$\text{Brier} = n^{-1} \sum_i (p_i - z_i)^2, \quad \text{LogLoss} = -n^{-1} \sum_i [z_i \log p_i + (1 - z_i) \log(1 - p_i)]. \quad (85)$$

Predicted diameter $\hat{d}_i$ is scored on genuinely alive records with observed DBH using mean signed error, MAE, RMSE, and central 50%, 80%, 90% draw intervals. Growth-error scores divide diameter errors by the actual measured interval. Aggregate held-out cohort counts, basal area, juvenile support and taxon shares are scored separately; observed alive records with missing DBH receive a separate unavailable-DBH status in aggregate truth scoring, preserving their observed living-status evidence. The subset/sample dependence of repeated cells and masks is retained in interpretation. Confidence intervals from latent reconstruction are conditional model intervals. Independent-sampling frequentist coverage requires a distinct sampling justification.

5.2 Alternative missingness families propagated to final queries

The sensitivity study retains baseline B0 reconstruction, then substitutes one of four predeclared unknown-fate hazard-shift mechanisms. For a cell c , Eq. (41) continues to apply with the corresponding latent shift η_c : B0 draws $\eta \sim N(0, .45^2)$ and $\eta_w \sim N(0, .7^2)$; B2 uses scales (.15, .25); B3 uses (.9, 1.1); B4 uses (.3, 1.2) and mean +.25 for the wet-candidate shift. B1 uses zero random shifts and a deterministic additive log-hazard structure

$$\eta_c^{B1} = a_{k(c)} + b_{z(c)} + g_{s(c)}, \quad (86)$$

with $a = (.1, 0, -.1, 0)$, $b = (.1, 0, 0, -.1)$ and $g = (0, 0, 0, .2)$. These are alternative *latent-fate reconstruction families*; B1 is MAR-like with respect to its selected observed covariates, and none is a fully identified model of the acquisition process itself.

The experiment samples from the finite stipulated family

$$\mathcal{B}_{\text{tested}} = \{B_0, B_1, B_2, B_3, B_4\} \subseteq \mathcal{B}_{\text{plausible}}, \quad (87)$$

with empirical coverage restricted to the declared family $\mathcal{B}_{\text{tested}}$. Hence the reported sensitivity is quantified over the explicitly explored family; data-acquisition missingness outside it remains unidentified. The 31 association-branch priors and other reconstruction mechanisms remain fixed.

For each B0–B4 family, the full chain is rerun with 128 state draws, one parameter draw per state, two growth variants, 256 future histories per outer unit, four scenarios, and 5-, 10-, 20-year horizons. The same declared semantics give 4,860 phase specifications per family. Random stream names and indices are shared where appropriate; variable draw consumption can differ, and the reported comparisons use shared stream identifiers with explicitly limited common-random-number coupling.

For paired conditional units u and specification q , the reported disagreement relative to B0 is

$$D(B_j, B_0) = \frac{1}{|\mathcal{U}||\mathcal{Q}|} \sum_{u,q} \mathbf{1}\{H_{u,q}^{\text{point}}(B_j) \neq H_{u,q}^{\text{point}}(B_0)\}, \quad (88)$$

with corresponding mean viable-mass differences reported *before* thresholding. The statistic is a finite-design discrepancy across declared mechanism variants; its interpretation is restricted to the finite-design discrepancy, with posterior mechanism probabilities and ecological reversal probabilities left unestimated. Each phase row retains numerical-resolution brackets in addition to the point label.

5.3 Census-entry identity audit

Each of 6,992 E1-only living stems with measured DBH is compared with the E0 stem and plant tables, both available tags, taxon, position, quadrat, measurement date/height, and proxy support. The E0 plant table contains 83,813 entries. A corresponding published E1 plant table is unavailable in HF253 revision 6; E1 plant grouping is consequently a derived classification. The audit retains the classes CERTAIN, POSSIBLE_ENTRY, POSSIBLE_E0_MISS, IDENTITY_AMBIGUITY, SAMPLING_FRAME, and UNRESOLVED, with explicit row-level reasons.

Diagnostic rates are formed from a class-membership sensitivity parameter $\omega_k \in [0, 1]$:

$$r^{(\omega)} = \frac{1}{6} \sum_k \omega_k N_k. \quad (89)$$

The lower bound counts independently certain entries; the upper bound treats all 6,992 observed E1-only candidates as entrants. The stipulated intermediate weights are 1 for certain, 0.5 for possible entry, 0.1 for possible E0 miss, 0.25 for identity ambiguity and unresolved, and 0 for sampling-frame difference. These stipulated weights define a sensitivity design; empirical calibration of entry-class membership probabilities remains open. The model includes extrapolated wet-proxy entry rates, so the diagnostic extrapolation extends beyond directly observed entry support.

5.4 Discriminative known-truth synthetic benchmark

The broad-range synthetic validation uses a factorial set of known generating states, conditional demographic laws and separately simulated observation packets. The world design and truth bank are fixed before scoring the estimator.

The broad-range world design uses the full factorial grid

$$3 \text{ structural levels} \times 3 \text{ regenerative levels} \times 3 \text{ pressure levels} \times 3 \text{ recovery levels} \times 4 \text{ replicates} = 324. \quad (90)$$

Latent state basal-area targets are $(.55, .78, 1.10)B_0$ and juvenile targets $(.20, .45, .90)J_0$, each perturbed by predeclared small lognormal variation (log SD .03). Adult cohort diameters are proportionally rescaled to obtain the target basal area after juvenile contribution, subject to $[1, 300]$ cm bounds. The annual pressure hazard levels are $(.005, .03, .12)$ and regeneration intensities are related to $(.025, .008, .001)J_0$; log-prior widths for hazard and entry parameter draws are .15 and .20, respectively. The experiment exposes the discrete pressure and recovery context labels as imposed design covariates, retaining actual latent states and numeric parameters exclusively in the truth generator.

A synthetic observation packet is formed by independently thinning each latent cohort count with probability 0.7 and perturbing a retained RMS diameter by $N(0, .25^2)$ cm, with clipping to model

bounds. The estimator’s admissible input type consists of counts, available diameters, observation probabilities, imposed public context labels, a fixed E0 anchor, and model configuration. Its observation module rejects latent-world and truth-parameter objects. The broad-range estimator draws 64 observation-conditioned states/parameter configurations and simulates 256 futures per draw, and evaluates the declared Health predicate in each draw. Its reported Health fraction is a *finite observation-conditioned design fraction*.

The truth generator independently enumerates a 4,096-history finite reference bank for each world and mechanism. Truth evaluation uses $t = 5$, $\alpha = .75$, $\beta = .4$, $\tau = 2$, $\theta = .75$ and $\rho = .5$. The finite truth mass is an approximation to the underlying stochastic law; a separate Wilson status reports its numerical decision resolution. The full truth census is archived before estimator performance is evaluated. All registered synthetic worlds remain in the evaluation roster regardless of true or estimated Health.

Three separately evaluated mechanism suites are specified. The correct-model suite shares the demographic kernel family and observation operator. The first misspecification adds 0.12 annual hemlock hazard in truth, absent from the estimator. The second adds independent annual path-level lognormal multiplicative entry intensity with log SD 1.2 and unit mean, absent from the estimator. The 324 initial states are reused across the three mechanisms as paired worlds, with statistical independence assessed at the initial-state level. Discrimination metrics use $\hat{p}_H \geq .5$ as the descriptive predicted-positive rule: sensitivity, specificity, balanced accuracy, Brier score, confusion matrix and truth-versus-estimate viable-mass MAE. Estimator mass-interval coverage and truth-bank resolution are reported separately from classification performance.

5.5 Adequacy-boundary world design and observation-only inference

Two disjoint pilot designs characterized how specified structural and regenerative parameters populate neighborhoods of the adequacy threshold. Their trajectories were excluded from the final evaluation; the complete final world roster was fixed before truth scoring and estimator outcomes. Every generated world, including six reference-uncertain cases, remains in the analysis.

The 81-world confirmatory roster comprises 54 Q1 targets (nine pressure/recovery contexts at six predeclared targets), 18 Q2 targets (three recovery contexts at six targets under pressure level 2), and nine present-realization controls. The predeclared target masses are (.675, .715, .745, .755, .785, .825). Its imposed world parameters declare controlled experimental conditions; calibration to Harvard Forest ecology remains a separate requirement. Define the signed Q1 and Q2 margins against $\theta = 0.75$ by

$$\delta_1 = m_V - 0.75, \quad \delta_2 = m_R - 0.75, \quad (91)$$

where m_V and m_R are masses under each world and its own declared generating law. The Q2 margin depends on the reserve-qualified mass even if the total continuing mass is far above threshold. We separately tag failures of present **Realizes**, continuation adequacy, and reserve adequacy. The target range is $m \in [.65, .85]$, with predeclared signed bins $[-.10, -.05]$, $[-.05, -.02]$, $[-.02, 0]$, $[0, .02]$, $[(.02, .05]$, $[(.05, .10]$. Generated truth determines actual stratum membership; the entire roster is retained.

For each world and each of the baseline, omitted-hemlock-hazard, and annual-entry-overdispersion suites, the generator produces a maximum $K_{\text{truth}} = 65,536$ -path reference bank. Earlier paired prefixes 4096 and 16,384 permit a convergence comparison under a fixed maximum-bank sampling

design. Reference response counts and Wilson-based uncertainty labels are stored before estimator results; reference uncertainty is retained in final tables. A world with unresolved reference truth is omitted from *certified-truth confusion-score denominators*, with its original roster, estimators and finite-bank answer retained. The reference bank is an exact census of its finite sampled histories and a Monte Carlo approximation to the declared generating stochastic law.¹

Synthetic observation kernel and inverse packet estimator. For a latent synthetic $X = (n_c, d_c)_{c=0}^{63}$, the synthetic observed packet $\tilde{Y} = ((\tilde{n}_c, \tilde{d}_c, p_c))_c$ is drawn from

$$\tilde{n}_c \mid X \sim \text{Binomial}(n_c, p_c), \quad p_c = .7, \quad (92)$$

$$E_c \mid X \sim N(0, .25^2), \quad \tilde{d}_c = \begin{cases} \text{clip}_{[1,300]}(d_c + E_c), & \tilde{n}_c > 0, \\ \perp, & \tilde{n}_c = 0. \end{cases} \quad (93)$$

The independent product of the cellwise laws, followed by the missing-diameter rule, defines $O(d\tilde{y} \mid X)$. Its inputs and outputs are synthetic population/cohort observations and is distinguished by its synthetic input type and sampled observation law from the archival HF253 census and the capacity observer o .

For a supplied strict observation packet, the estimator fixes the public E0 anchors (n_c^0, d_c^0) and draws a latent count intensity and unobserved count by

$$a_c = 16 / \max\{n_c^0, 1\}, \quad (94)$$

$$L_c \mid \tilde{Y} \sim \text{Gamma}\left(16 + \tilde{n}_c, \frac{1}{a_c + p_c}\right), \quad (95)$$

$$n_c^* = \tilde{n}_c + U_c, \quad U_c \mid L_c \sim \text{Poisson}((1 - p_c)L_c). \quad (96)$$

For a present diameter, let $v_0 = 1^2$, $v_1 = .25^2$, $\iota_c = \mathbf{1}\{\tilde{d}_c \neq \perp\}$ and write

$$v_c^* = (v_0^{-1} + \iota_c v_1^{-1})^{-1}, \quad (97)$$

$$\mu_c^* = v_c^* (d_c^0 v_0^{-1} + \iota_c \tilde{d}_c v_1^{-1}), \quad (98)$$

$$d_c^* = \text{clip}_{[1,300]}(Z_c), \quad Z_c \sim N(\mu_c^*, v_c^*). \quad (99)$$

For a label-only negative control the packet has $p_c = 0$, $\tilde{n}_c = 0$, and $\tilde{d}_c = \perp$ for all cells; the estimator then samples from the declared public-anchor prior. These Gamma–Poisson and normal precision rules define the synthetic observation-conditioned estimator, a separate model component from the empirical reconstruction kernel of Eq. (50).

5.6 Observation-only estimator and the role of context labels

The estimator receives a synthetic thinned/noisy observation packet $\tilde{Y}$ and permitted public context markers. The truth state, truth parameters, future history identities, truth masses, and true Health outcomes are excluded from its interface. The boundary information-policy evaluation uses $N_e = 32$ state/parameter draws, each with $K_e = 256$ conditional futures; it computes a design fraction of

¹Computational counterpart: `HFD04/config/world_design.json`; `HFD04/scripts/h04_boundary.py`; `boundary_validation/truth_prefixes.csv`; `boundary_validation/pre_estimator_census.json`.

draw-specific Health propositions. The broad-range and boundary experiments use their explicitly stated and distinct estimator resolutions.

The synthetic pressure label $p \in \{0, 1, 2\}$ specifies a prior anchor for annual hazard,

$$\bar{h}_p = (0.005, 0.03, 0.12)_p, \quad h_c = \bar{h}_p \exp(Z_{h,c} - .15^2/2), \quad Z_{h,c} \sim N(0, .15^2).$$

The recovery label $\ell \in \{0, 1, 2\}$ supplies a baseline-relative total entry anchor $r_\ell = (.025, .008, .001)_\ell$, allocated to cell c through

$$s_c = \frac{\lambda_c^{\text{fitted}} + 1/64}{\sum_b (\lambda_b^{\text{fitted}} + 1/64)}, \quad \lambda_c = r_\ell J_0 s_c \exp(Z_{\lambda,c} - .20^2/2), \quad Z_{\lambda,c} \sim N(0, .20^2). \quad (100)$$

These are declared experimental prior mappings; a known label supplies meaningful information even though the true per-cell hazards and generating entry rates remain concealed from the estimator. The broad-range $3^4 \times 4 = 324$ factorial roster has equal representation of nine (p, ℓ) combinations. The boundary roster has counts $(7, 7, 7, 7, 7, 7, 13, 13, 13)$ in pressure-major, recovery-minor order. The distinction matters for unknown-context mixtures.

Four information policies are evaluated on paired worlds and observation packets: $B0$ retains both labels; $B1$ coarsens pressure into $\{0, 1\}$ versus $\{2\}$ and recovery into $\{0\}$ versus $\{1, 2\}$, with uniform design mixing within each reported group; $B2$ withholds both labels and mixes over all nine imposed contexts with uniform $1/9$ design weights; and $B3$ supplies labels but replaces informative state measurements by missing markers and uses the public $E0$ state prior. The class $B2$ estimates an unknown-realized-context predictive problem; the formal proposition $H_S(x; d, t)$ remains indexed by a known d in its ordinary use. No context mixture is identified as an ecological frequency or posterior prior.

A separate registered sensitivity analysis uses the marginal *design distribution* for the boundary roster by using $w_{p\ell} = n_{p\ell}/81$, where $n_{p\ell}$ is 7 for the first six combinations and 13 for the last three. The roster-weighted and uniform-nine mixtures are reported separately, with independent PCG64 streams and complete world rosters. Differences between the two unknown-context mixtures contain both prior-design and finite-draw variation. The roster weights were computed from the complete designed world roster independently of the scored outcomes.²

5.7 Four-way stagewise oracle experiment

The same 81 boundary worlds are evaluated under a factorial information intervention. Let X^*, ϑ^* denote the latent generating state and demographic parameters, and let $\hat{X}, \hat{\vartheta}$ denote information-restricted estimates from the synthetic observation packet and disclosed labels. Define

	state supplied	parameter supplied
E_{00}	$\hat{X}$	$\hat{\vartheta}$
E_{10}	X^*	$\hat{\vartheta}$
E_{01}	$\hat{X}$	ϑ^*
E_{11}	X^*	ϑ^*

(101)

²Computational counterpart: `HFD04/config/context_prior_audit.json`; `HFD04/scripts/h04_estimator.py`; `HFD04/scripts/h04_context_prior_estimator.py`; `context_ablation/prior_reconciled_comparison.csv`.

All four roles use the same declared continuation and adequacy semantics and a separately specified history generator. The observation-only interface rejects oracle payloads. Future numerical paths are generated with fixed prefixes and replicate-source panels; supplying oracle inputs does not eliminate stochastic future variation or a misspecified truth/estimator kernel. The four roles are compared on each certified reference truth and margin stratum. With target error functional L_{ab} , a signed factorial interaction is

$$I_L = L_{11} - L_{10} - L_{01} + L_{00}. \quad (102)$$

Signed main contrasts and interaction are reported as designed interventions on the synthetic inverse model, with no additive ecological error decomposition assumed. Between-world uncertainty respects the constructed strata; empty truth groups retain undefined rates.

Path-resolved test of the selected one-year contrast. The selected comparison fixes state 205, predictive parameter replicate 1, D0, Gamma growth, $(t, \alpha, \beta, \tau) = (1, 0.5, 0.8, 0)$ and the original frozen 4,096-path future streams. For each conditional bank, every simulated trajectory retains its terminal juvenile count, terminal basal area, and model-admissibility flag. The indicator identities in Eq. (75) assign every history to exactly one of five categories. An independent scalar implementation checks those labels against original origin-state summaries and the recorded viable totals; the full overlapping-cause table is also retained. The two banks are separately generated conditional response laws, so identical named stream identities are not interpreted as a pathwise causal pairing. The observed classification is presented alongside the capacity contrast in Sec. 6.4.

5.8 Matched external adult-mortality component test

We evaluate the fixed earlier-census hazard distribution against HF453 adult stem statuses recorded in 2021–2024 [6]. A unique stable stem tag, compatible recorded species, earlier adult DBH support (≥ 10 cm), observed 2021 alive status, and eligible later endpoints are required. Stem/tree collisions, mismatched identifiers, baseline dead or absent observations, nonadult earlier stems, secure dead-to-alive reversals and unsupported visits remain in a source-level exclusion ledger. The baseline is *left-truncated*: a stem must survive to be observed alive in 2021; the test consequently estimates mortality conditional on 2021 survival, with no inference about 2020–2021 deaths.

Let t_{i0} be the risk-set baseline and t_{iy} the selected annual or cumulative endpoint. The fitted hazard-grid distribution $\hat{\pi}_{c(i)}$ from Eq. (34) defines the conditional prediction

$$p_i^{\text{death}}(y) = 1 - \sum_{g=1}^{256} \hat{\pi}_{c(i)}(h_g) e^{-h_g \Delta_{i,y}}, \quad \Delta_{i,y} = t_{iy} - t_{i0}. \quad (103)$$

Joint predictive count law and shared-hazard uncertainty. The per-stem probabilities in Eq. (103) determine expected totals. Predictive uncertainty for the total number of deaths also includes uncertainty shared by stems assigned to the same fitted hazard cell. For the fixed, strictly scored risk set $\mathcal{I}$, let $\mathcal{I}_c$ be the stems assigned to cell c , and define

$$q_i(h) = 1 - e^{-h \Delta_{i,y}}, \quad A_c(h) = \sum_{i \in \mathcal{I}_c} q_i(h), \quad V_c(h) = \sum_{i \in \mathcal{I}_c} q_i(h)(1 - q_i(h)). \quad (104)$$

Under the implemented design law, the hazard H_c is drawn once per cell from $\hat{\pi}_c$ and is shared by the conditionally independent stem outcomes in that cell. For the declared independent sampling of

cell hazards, total predictive deaths $D = \sum_c D_c$ have

$$\begin{aligned}\mathbb{E}(D \mid Y_{\text{fit}}, \mathcal{I}) &= \sum_c \mathbb{E}_{\hat{\pi}_c} \{A_c(H_c)\}, \\ \text{Var}(D \mid Y_{\text{fit}}, \mathcal{I}) &= \sum_c \left[\mathbb{E}_{\hat{\pi}_c} \{V_c(H_c)\} + \text{Var}_{\hat{\pi}_c} \{A_c(H_c)\} \right].\end{aligned}\tag{105}$$

The first variance contribution is the conditional Bernoulli event variation; the second is the uncertainty in the cell's expected death count induced by sharing one sampled hazard. If a different joint parameter law couples cells, corresponding cross-cell covariance terms are required. The fitted $\hat{\pi}_c$ is a stipulated, likelihood-tempered parameter law; Eq. (105) is therefore a *model-based predictive uncertainty decomposition*, with the risk set and source-selection convention held fixed.

The fitted law further specifies the total-death predictive distribution. Conditional on each vector of sampled cell hazards, stem death indicators are independent Bernoulli variables. Marginally the cell count is a *mixture of Poisson–binomial laws*, with probability-generating function

$$\mathcal{P}_c(z) = \sum_{g=1}^{256} \hat{\pi}_c(h_g) \prod_{i \in \mathcal{I}_c} [(1 - q_i(h_g)) + q_i(h_g)z], \quad \mathcal{P}_D(z) = \prod_c \mathcal{P}_c(z).\tag{106}$$

The product over cells uses the study's independent-cell parameter law; the shared hazard is sampled once per cell. Equation (106) permits heterogeneous $\Delta_{i,y}$ and hence conditional Poisson–binomial cell counts. For the executed *strict cumulative 2021–2024 predictive-distribution test*, the frozen July-anniversary convention assigns exactly $\Delta_{i,2024} = 3$ years to *every* stem in its 21,576-stem matched risk set. Each Binomial sampling group is one occupied original taxon–stratum–size hazard cell $\mathcal{I}_c$ (among the 64 possible cells); all $n_c = |\mathcal{I}_c|$ stems in that cell therefore share both the sampled hazard H_c and the conditional death probability

$$q_c(h) = 1 - e^{-3h}, \quad D_c \mid H_c = h, \mathcal{I} \sim \text{Binomial}(n_c, q_c(h)), \quad D_G = \sum_{c \in G} D_c.\tag{107}$$

The independent count-level verification draws one H_c from $\hat{\pi}_c$ for each occupied cell, then one conditional Binomial D_c under Eq. (107), and sums these counts over the relevant hemlock, nonhemlock or all-adult cells. It thereby samples the same hierarchical law as the individual-event Bernoulli implementation. No single Binomial distribution is asserted for an entire taxon group, and no new hazard is drawn independently for each stem. The common three-year exposure is an explicitly declared approximation for this endpoint; the stem-specific formulation in Eq. (106) remains necessary for genuinely heterogeneous exposures and the separately reported anniversary-time sensitivity.

A Monte Carlo evaluation of this defined distribution draws $H_c^{(b)} \sim \hat{\pi}_c$ once per cell, then draws $Z_i^{(b)} \mid H_{c(i)}^{(b)} \sim \text{Bernoulli}\{q_i(H_{c(i)}^{(b)})\}$ for every eligible stem. The replicate count and empirical upper-tail statistic are

$$D^{(b)} = \sum_{i \in \mathcal{I}} Z_i^{(b)}, \quad \hat{p}_{\geq} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{D^{(b)} \geq D_{\text{obs}}\}.\tag{108}$$

The model-based predictive question concerns the location of each observed count relative to the corresponding distribution: report the predictive median, central 95% equal-tail interval and $\Pr(D \geq D_{\text{obs}} \mid Y_{\text{fit}}, \mathcal{I})$, with Monte Carlo uncertainty if simulation is used. Hemlock and other-taxon totals must be computed from the *same* jointly sampled cells and exact primary scored risk set,

preserving their dependence. The spatial-cluster resampling interval addresses uncertainty of the observed-minus-predicted *mean residual* conditional on observed source support. We evaluate the joint predictive law on the strict matched adult risk set with 262,144 individual-event replicate draws using one sampled hazard per occupied cell. An independent sampler draws conditional Binomial counts *within each occupied hazard cell*, using the same common three-year exposure and one sampled shared cell hazard as Eq. (107); adult-group totals are sums of these cell counts. The independent implementation was checked against the stem-event sampler, the exact finite mixture law, and a small-cell full event enumeration. Since extreme upper-tail events can be absent from finite Monte Carlo replicates, we separately evaluate predictive probability masses by positive mode-centered recurrence and direct convolution, independently checked by a loggamma/direct method. Their agreement establishes the reported rare-tail probabilities at numerical precision substantially finer than the simulation-only zero-hit upper bound. Predictive and cluster-bootstrap intervals are reported separately in Sec. 6.5.

Shared cell hazard, epistemic mixing, and survivor selection. The discrete distribution $\hat{\pi}_c$ describes uncertainty about a single fitted cell-level hazard, conditional on the earlier-census evidence. Its predictive mixture in Eq. (103) is not a fitted distribution of persistent, individual-level frailties. Conditional on a sampled shared h_c , the model applies stem-specific elapsed times and conditionally independent survival events; integrating the shared h_c can induce dependence among marginal event indicators. If S_{2021} denotes survival into the observed 2021 adult risk set, a fully survivor-updated shared-hazard law would formally satisfy

$$\Pi_c(dh \mid Y_{\text{fit}}, S_{2021}) \propto \Pr(S_{2021} \mid h, Y_{\text{fit}}) \Pi_c(dh \mid Y_{\text{fit}}), \quad (109)$$

provided that the selection likelihood and any overlap with the fitting data are identified. The reported retrospective component test uses the *frozen earlier-fitted* $\hat{\pi}_c$ in Eq. (103), with no additional 2021 survivor-selection likelihood or external-outcome update. Its tested target is mortality of eligible stems observed alive in 2021, and the baseline excludes unobserved earlier deaths. A persistent stem-frailty model would require a separate individual random-effect law and its own selection adjustment; neither is estimated here. The observed mortality discrepancy is therefore a diagnostic of this precisely defined predictive mixture on its selected support.

Year-only visit precision is treated as an anniversary-time approximation with a ± 30 -day prediction-sensitivity calculation. Recorded A/AU categories count as alive and DC/DS as secure dead in the primary test; DN (dead/not found) is excluded from the primary secure-death denominator and evaluated separately under DN-as-dead and DN-as-alive sensitivity conventions. No missing visit date or death date is imputed as an observation. For eligible endpoint labels $y_i \in \{0, 1\}$, the reported scores are

$$\text{Brier} = n^{-1} \sum_i (p_i - y_i)^2, \quad \text{LogLoss} = -n^{-1} \sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]. \quad (110)$$

Predicted and observed event totals, calibration bins, hemlock versus nonhemlock outcomes, taxon, earlier DBH, stratum/coverage proxy and recorded-damage subsets are reported with matched denominators. Whole-stem-history and quadrat-block descriptive resampling retain longitudinal clustering. Later-year outcomes are withheld from fit and tuning. This is retrospective *component validation*; it does not constitute a previously issued prospective forecast or evaluation of the complete forest Health predicate.

5.9 Independent seedling support and entry identifiability

HF355 supplies repeated below-1 cm seedling observations within 1 m^2 sampled subplots [7]. Matchable seedling identifiers, observed visits, actual date intervals, secure status changes and explicit graduated/first-recruit codes form the eligible descriptive transition carrier. Reversed secure statuses and unavailable temporal support are excluded with reasons. First observation establishes a left-observation boundary; germination date remains unobserved. Initially graduated stems provide left-censored graduation records. The data support descriptive conditional seedling survival and field-coded recruitment interpretation. Their spatial/diameter support does not identify the whole-plot 1–10 cm stem-entry process Λ_c .

The independent identity audit of the later-only 6,992 candidate stems similarly asks which recorded variables distinguish an actual 1 cm crossing from an E0 miss, a new stem on an existing plant, a tag/identity change or a sampling-frame difference. Observationally equivalent latent histories yield a noninjective map from biological entry histories to the available two-census record. The inference therefore retains candidate-entry sensitivity weights as defined assumptions and never interprets them as fitted latent-category probabilities.

5.10 Query-specific juvenile membership at the 10-cm threshold

For stem diameters d_i , the juvenile support needed by $R_{\alpha,\beta}$ and Q2 is $J = \sum_i \mathbf{1}\{1 \leq d_i < 10\text{ cm}\}$. The original aggregate state retains per-cell count n_c and RMS diameter d_c^{RMS} , where

$$(d_c^{\text{RMS}})^2 = \frac{1}{n_c} \sum_{i \in c} d_i^2, \quad J_{\text{RMS}} = \sum_c n_c \mathbf{1}\{d_c^{\text{RMS}} < 10\}. \quad (111)$$

The first identity conserves within-cell basal area; the second classifies an entire cohort on a nonlinear threshold. The size-audit compares J_{stem} and J_{RMS} among 53,611 securely observed living E1 stems with documented dates and size fields. Direct normal perturbations at diameter SD 0, 0.1, 0.25, 0.5, 1 cm form separately labeled methodological stress tests. Existing physically sanitized M0–M3 real-data masks also evaluate juvenile membership, absolute count bias, and $\Delta(J/J_0)$ at the observed E1 support.

The enriched representation divides each cohort into below-10 and at-least-10-cm components using size profiles learned from *visible* E1 records, with explicit fallbacks where information is insufficient. For a reconstructed origin cohort, it chooses integer subcounts and rescales the two subdiameters to conserve total count and second moment,

$$n_c = n_{c,<10} + n_{c,\geq 10}, \quad n_c(d_c^{\text{RMS}})^2 = n_{c,<10}d_{c,<10}^2 + n_{c,\geq 10}d_{c,\geq 10}^2. \quad (112)$$

The projection preserves initial count and basal area to numerical tolerance. Juvenile support J can change under projection, and a future stochastic-kernel intertwining relation remains unproved. Matched seeds make the finite-design comparison paired, while changed random-array dimensions prevent an assertion of exact identical underlying uniforms. All inferred size substructure retains its imputation provenance.

To exhibit the query obstruction independently of any empirical estimator, compare two 100-stem fine states: in A all diameters are 10 cm, and in B 50 are 9 cm and 50 are $\sqrt{119} \approx 10.9087$ cm. Both have count 100 and RMS diameter 10 cm, hence equal basal area $100\pi(10)^2/40000\text{ m}^2$. Yet $J_A = 0$

and $J_B = 50$. A juvenile-sensitive realization/response query can consequently take opposite truth values on the same coarse representation. This is an explicit finite counterexample to deterministic query factorization through $(n, d^{\text{RMS}}, \text{BA})$.

Proposition 5.1 (Juvenile-query non-factorization). *Let π_{RMS} send a fine stem-diameter multiset to its cohort count, RMS diameter and resulting basal area. For the declared juvenile-sensitive query $q_J(X) = \mathbf{1}\{J(X) \geq 1\}$, the coarse RMS image admits no map $\bar{q}_J$ satisfying $q_J = \bar{q}_J \circ \pi_{\text{RMS}}$ on all fine states.*

Proof. Take X_A to be 100 stems each of diameter 10 cm and X_B to be 50 stems of diameter 9 cm and 50 of diameter $\sqrt{119}$ cm. Both have $\sum_i d_i^2 = 10,000$ and $n = 100$, so $\pi_{\text{RMS}}(X_A) = \pi_{\text{RMS}}(X_B)$. Meanwhile $J(X_A) = 0$ and $J(X_B) = 50$, hence $q_J(X_A) = 0 \neq 1 = q_J(X_B)$. The two distinct query values establish failure of factorization through their common coarse image. $\square$

For requirement configurations in which the reserve-qualified Q2 answer differs between these two fine histories, the same witness shows Q2 non-factorization. The exact result concerns information retained by π_{RMS} ; the 256-path enrichment comparison changes a conditional stochastic implementation with transition-kernel commutation remaining an independent unproved condition. It supports the representation-sufficiency analysis; ecological generalization requires separate licensing.³

5.11 Effective-count cap sensitivity in the fitted demographic law

The fitted model tempers cell c 's unequal-interval hazard likelihood by $\kappa_c = \min(1, n_{\text{cap}}/n_c^{\text{fit}})$ to limit the nominal independent information of spatially correlated stems. The sensitivity study refits from the identical E0/E1 records with

$$n_{\text{cap}} \in \{50, 100, 200, 500\}, \quad p_c(h \mid D_c, n_{\text{cap}}) \propto p_c(h) \exp\{\kappa_c \ell_c(h)\}, \quad (113)$$

where $\ell_c(h)$ is the previously specified live/dead unequal-exposure log likelihood, the hazard grid contains 256 points, and the Beta annual-survival prior retains strength 32. Growth standard error is

$$\text{SE}_c = \frac{\max(0.02, \hat{\sigma}_c)}{\sqrt{\min\{\max(n_c^{\text{growth}}, 1), n_{\text{cap}}\}}}.$$

Candidate entry Gamma shape also has a cap through $\min(6\bar{\lambda}_c + 1, n_{\text{cap}})$. The cap changes the model's effective information assumptions; it is an imposed tempering parameter whose relation to the actual spatial-dependence effective sample size remains unidentified.

For each cap the original fitting and reconstruction code is used in isolated module globals, then paired origin states and new futures are generated for four scenarios, 5- and 10-year horizons, Q1/Q2, and the selected semantic cells. The reported small panel has 16 state draws, two model variants and 256 inner paths. A separate M0 held-out calibration uses 64 reconstruction draws per cap. The cap-200 reference result remains unchanged, and every cap value is retained independent of E2 aggregate agreement.⁴

³ Computational counterpart:

HFD04/juvenile_boundary/projection_counterexample.json; HFD04/scripts/h04_juvenile.py.

⁴ Computational counterpart:

HFD04/effective_count/fit_contract.json; HFD04/scripts/h04_cap.py; effective_count/propagation.csv.

5.12 Finite-bank decisions and stochastic-kernel resolution

Given a bank of K unconditional candidate histories, count $N_V = \#\{\gamma : \text{Lawful}_G(\gamma) \wedge \text{Continues}(\gamma)\}$ and $N_R = \#\{\gamma : \text{Lawful}_G(\gamma) \wedge \text{Continues}(\gamma) \wedge r_J(\gamma) \geq \rho\}$. The exact finite-bank masses are $\hat{m}_V = N_V/K$ and $\hat{m}_R = N_R/K$. Finite Q1 and Q2 answers use inclusive thresholds as in Eq. (77). With Bernoulli count x , the approximate pointwise 95% Wilson center and half-width are

$$C = \frac{\hat{p} + z^2/(2K)}{1 + z^2/K}, \quad W = \frac{z}{1 + z^2/K} \sqrt{\frac{\hat{p}(1 - \hat{p})}{K} + \frac{z^2}{4K^2}}, \quad z = \Phi^{-1}(0.975). \quad (114)$$

Use the clipped interval $[C - W, C + W]$ for N_V in Q1 and for N_R alone in Q2. The reserve event is a subset of viability, so the second interval needed by a redundant two-inequality formulation contributes no additional Boolean constraint. For a present-realizing state and applicable lower/upper bounds (L, U) , define

$$\text{MCstatus} = \begin{cases} \text{FALSE}, & \text{Realizes}(X_0) = 0 \text{ or } U < \theta, \\ \text{TRUE}, & \text{Realizes}(X_0) = 1 \text{ and } L > \theta, \\ \text{MC_UNRESOLVED}, & \text{otherwise.} \end{cases} \quad (115)$$

At $\theta = 0$, adequacy holds algebraically and present realization determines Health. At $\theta = 1$, an all-success finite bank may have exact finite Health TRUE while the stochastic law's mass-one claim remains uncertified. The interval is conditional on one generator and its i.i.d. path sampling assumptions; it gives no simultaneous coverage across all scenarios/queries or ecological model uncertainty. Original candidate weights are never renormalized by viability.

5.13 Correction-factorial, exposure and numerical-sensitivity designs

A five-identity factorial distinguishes a pooled-time, shape-floored reference computation (R_0); the mathematically equivalent single-reserve inference on those banks (R_1); the mean-preserving Gamma law with pooled exposure (R_2); individual-time reconstruction with the reference Gamma law (R_3); and the combined individual-time and mean-preserving Gamma system (R_4). These are *model sensitivity arms* evaluated on common declared inputs and separately indexed laws. The primary analysis uses R_4 ; R_0 – R_3 quantify the influence of numerical and modeling assumptions. Each has 256 origin-state draws, two predictive parameter draws per state, two growth variants, 64 futures per unit/scenario and the four scenarios and semantic grid in Eq. (79). The corresponding five-family missingness analysis uses 128 states, two growth variants and 256 inner histories; all finite responses retain their unconditional candidate denominators.

Conditional exposure effects are assessed with captured shared hazards, missingness shifts, diameter disturbances, entry and association draws. Individual Bernoulli sampling can still differ from pooled binomial sampling; paired random seeds alone do not assert identical coupling. Report expectations analytically and observed state/decision contrasts through independent paired numeric samples. For sampled extremes, additional prespecified prefixes through 4,096 paths distinguish targeted large capacity shifts from coarse finite-bank variation. The largest observed differences are diagnostics over the executed design, not general estimates of maximum ecological effect.

The model-variant families are also interrogated through moment tests of floor activation, all reached parameter identities, alternate effective-count caps, and conditional unknown-fate/entry

perturbations. The source and model identities of any re-evaluated reference truth are explicit. Revised boundary reference banks use 4,096 and 16,384 future paths; the separately identified 65,536-path reference from the boundary experiment is not silently treated as a revised-law reference. Synthetic observation packets do not contain individual dates, so the exposure-factor comparison is inapplicable to their inverse estimator.

Matched exposure intervention. The controlled comparison conditions on the observed status groups and their source dates, the same fitted hazards, sampled shared shifts, growth parameters, component-level size noise, association branches and unseen-entry draws. It changes the within-component treatment of the exposure durations through Eqs. (43) and (41). Sampled Bernoulli survivor realizations can differ, and predictive trajectories remain stochastic. The conditional analytic mean comparison isolates the duration operation; subsequent finite-path changes incorporate its stochastic realization. The first unrestricted random-stream comparison is retained only as a separate sensitivity check and is excluded from causal attribution to exposure alone.

5.14 Designed-world uncertainty and decision-margin analysis

The boundary study retains its complete 81-world roster and six originally unresolved maximum-bank truth states. For certified presently realizing worlds with signed query margin $|\delta_q| \leq 0.1$, define sensitivity $TP/(TP + FN)$, specificity $TN/(TN + FP)$, and balanced accuracy $BA = (Se + Sp)/2$. Wilson intervals quantify truth-class conditional binomial rates. A plug-in class-stratified parametric bootstrap uses

$$T_+^* \sim \text{Binomial}(n_+, \widehat{Se}), \quad T_-^* \sim \text{Binomial}(n_-, \widehat{Sp}), \quad BA^* = \frac{1}{2} \left(\frac{T_+^*}{n_+} + \frac{T_-^*}{n_-} \right). \quad (116)$$

These intervals are conditional on the deliberately constructed roster, and are distinct from confidence intervals for a real forest population. Every possible binary assignment of the six reference-uncertain cases is also enumerated, holding predictions fixed, to provide a separate truth-ambiguity bracket. Paired seed panels and constructed-stratum resampling distinguish observation/inference/future variability from reference-truth and ecological uncertainties.

6 Results

6.1 Two criteria for interpreting the study

We distinguish *specification-execution success*, established when the declared kernels, input identities, probability ledger, computations and numerical reporting agree with their mathematical definitions, from *empirical measurement success*, which further requires identified state and process variables, calibrated observation/predictive models, query-sufficient representations and sufficiently resolved decisions. The following results report both criteria for the precise population, synthetic design or observational support on which each was tested. All reported Health fractions are finite-design quantities indexed by a declared scenario and semantic requirement.

6.2 Prospective capacity and Health under the declared stochastic specification

Table 3 reports the primary joint individual-exposure/mean-preserving-Gamma implementation. The chosen illustrative requirement fixes $\alpha = 0.75$, $\beta = 0.4$, $\tau = 2$, $\theta = 0.75$ and reserve ratio $\rho = 0.5$. The table is a complete scenario-by-selected-horizon extract of the 1,024-unit design, with 64 simulated paths per unit. The reported masses average unconditional within-unit viable and reserve-qualified fractions; $\hat{H}_{Q1}$ and $\hat{H}_{Q2}$ are proportions of individually adequate, presently realizing design units. In general, the Health fraction is not a threshold applied to the ensemble-average mass. At 20 years, the response law and threshold criteria produce strong scenario differentiation and a

Scenario	Years	$\bar{m}_V$	$\bar{m}_R$	$\hat{H}_{Q1}$	$\hat{H}_{Q2}$	Unresolved Q1/Q2
D0	5	1.0000	1.0000	1.0000	1.0000	0/0
D1	5	1.0000	1.0000	1.0000	1.0000	0/0
D2	5	1.0000	1.0000	1.0000	1.0000	0/0
D3	5	0.9908	0.9908	0.9902	0.9902	0/0
D0	10	1.0000	1.0000	1.0000	1.0000	0/0
D1	10	0.9942	0.9942	0.9922	0.9922	0/0
D2	10	1.0000	1.0000	1.0000	1.0000	0/0
D3	10	0.0038	0.0019	0.0000	0.0000	0/0
D0	20	1.0000	1.0000	1.0000	1.0000	0/0
D1	20	0.2099	0.0779	0.1973	0.0723	16/8
D2	20	0.2815	0.2815	0.1318	0.1318	95/95
D3	20	0.0000	0.0000	0.0000	0.0000	0/0

Table 3: Primary conditional finite-design results from the declared individual-exposure, mean-preserving-Gamma study surface. The count of unresolved stochastic-kernel classifications is out of 1,024 units per query/scenario/horizon; finite-bank Health fractions themselves are exact for the generated paths. D0–D3 are stipulated generator scenarios. These values are neither ecological posterior probabilities nor operational forest Health diagnoses.

nontrivial difference between viability and reserve-qualified viability. Near-threshold unresolved decisions remain visible even when every finite-bank label is determined. The full experiment varies requirement values, growth variants and horizons beyond the displayed extract.

6.3 Identification and calibration: observed-state recovery

The tagged-stem record has substantial unresolved observation support. Across E0 and E1, 116,227 and 123,218 stem rows are present; E1 contains 37,577 `stem_gone` records and 61,985 rows without DBH. The reconstruction carries these as separately typed evidential states. Of 66,284 securely interpretable E0–E1 transitions used for the leave-observed-out experiment, four mask families with three replicates each yield the results in Table 4. Individual-diameter interval coverage is substantially below nominal on the held-out observed population. Aggregate cohort basal area behaves better because the fitted state representation preserves a second moment of diameter, while still discarding much of the individual-size distribution.

The fitted whole-plot latent state depends on observations and an unknown-fate mechanism, including common log-hazard shifts and the protocol-defined wet-candidate stratum. The observed-data masks test the reconstruction algorithm on records with observable truth. They do not identify

Mask family	Held-out stems / replicate	DBH 90% coverage	BA 90% coverage
M0, random	9,839–10,030	0.499–0.503	0.938
M1, taxon/size	13,112–13,136	0.468–0.474	0.970
M2, stratum	11,526–11,753	0.491–0.503	0.923
M3, protocol-pattern	8,118–8,151	0.551–0.560	0.947

Table 4: Conditional observed-data masking calibration. Diameter ranges aggregate the three replicates within each family; aggregate basal-area figures are the representative pooled 90% cohort-interval coverages for the corresponding family. The test population consists of securely recorded endpoints, with the genuinely unresolved population outside its direct calibration support.

the outcomes missing under the actual census process. The candidate entry audit separates this structural lack of identification from ordinary parameter uncertainty. Among 6,992 later-only measured living stems, no individual is certified as an unequivocal biological threshold entrant under the available two-census identity records. There are 6,888 possible threshold entries, four possible earlier misses, two identity ambiguities and 98 unresolved. An earlier-unrecorded eligible stem and an actual newly crossing stem can yield the same later record; the observation map is noninjective for biological entry.

The recorded later-only candidates have a concentrated size distribution (Table 5). The model’s annual birth rule in Eq. (39) maps the pooled candidate intensity, including contributions from larger observed diameter classes, into a cohort initialized at 1.7 cm. Excluding the 102 candidates

Recorded E1 diameter class	Candidate stems	Fraction of 6,992
$1 \leq D < 3$ cm	6,890	98.54%
$3 \leq D < 10$ cm	98	1.40%
$10 \leq D < 30$ cm	2	0.03%
$D \geq 30$ cm	2	0.03%
Total	6,992	100.00%

Table 5: Observed diameters of later-only living stem records, classified at the cohort size edges. These are first-appearance observations, not independently certified biological births or 1-cm threshold-crossing events.

at least 3 cm from the recorded candidate-input sum changes the nominal $e/6$ input by 17 records per year across the observed support. This 1.46% difference applies to the unweighted recorded candidate count. The sampled Gamma intensities, wet-proxy borrowing rule and nonlinear future decision mechanism have separate responses to a restricted refit. The present analysis does not assign an uncomputed future-capacity effect to that restriction. Even the 6,890 smaller candidates remain biologically unidentified. The corresponding birth-event mapping is therefore an explicit structural model assumption, with its predictive sensitivity left as a separate unestimated quantity.

6.4 Individual exposure and fitted growth: mathematical and numerical behavior

The individual-date inventory contains 98,018 dated reconstruction-component records. Direct evaluation of Eq. (46) establishes a nonnegative conditional survival-count expectation discrepancy from replacing heterogeneous stem exposures by the cell mean. Independent Bernoulli tests and

exact Poisson-binomial distribution calculations verify the corresponding count law. With shared reconstruction nuisance draws held fixed, the individual-time state yields a mean change of +26.34 living stems and +24.38 juvenile proxy stems, and a mean basal-area change of +0.0213 m² relative to pooled-time reconstruction. The largest sampled absolute differences are 471 stems, 465 juveniles and 5.49 m² in basal area, respectively. Consequently, a small mean change does not bound the local structural effect.

The growth-law audit identifies 12 positive shape-floor activations among 131,072 primary ancestral cell/scenario laws, arising from three sampled parameter identities. The floored reference shape-scale law deviates materially from its nominal mean in those cases. For example, one $\mu' = 4.23206 \times 10^{-5}$ had reference-law actual mean 3.186235×10^{-4} , while the mean-preserving law restores $\mathbb{E}G = 4.23206 \times 10^{-5}$. The reached-law impact panels and complete executed primary finite-design surfaces yield unchanged capacity and Boolean answers for the isolated Gamma-scale correction. That numerical finding coexists with a verified conditional moment defect in the alternative law; equality of continuous stochastic processes is not inferred.

The reserve requirement in Eq. (77) is exactly determined by $m_R \geq \theta$, by inclusion of reserve-qualified histories in viable histories. Independent count audits cover 4,212,549 finite evaluations with unchanged exact finite-bank Health labels. Using one pointwise 95% interval for the reserve count using the mathematically sufficient single component interval resolves 371 additional weighted evaluations in the retained retrospective inference census (158 false and 213 true); the underlying histories and capacities remain fixed.

Across the complete comparison, individual-time reconstruction changes some conditional Health decisions. Maximum observed semantic-cell disagreement is 0.06348 in the primary design and 0.04297 in the missingness analysis. One protocol-selected local contrast remains large at higher path counts. It uses D0, a one-year horizon, $\alpha = 0.5$, $\beta = 0.8$, $\tau = 0$, state identity 205 and parameter replicate 1. Table 6 records the two reconstructed origins and the 4,096-path capacity results under the shared fitted parameters and registered nuisance-draw control. The $0.8J_0$ threshold

	Pooled component exposure	Individual stem exposure
Living origin stems	90,431	90,690
Origin juvenile proxy stems	66,600	66,900
Origin basal area (m ²)	1,516.947	1,515.571
Model-lawful and continuing futures	43	3,959
Model-nonviable futures	4,053	137
Unrenormalized viable mass	0.01050	0.96655

Table 6: Selected one-year conditional exposure sensitivity, $K = 4,096$ futures in each bank. Nonviable counts are $K - n_V$; independent path-level replay assigns their complete disjoint terminal failure categories in Table 7. Both initial states satisfy the declared realization requirement. This is a deliberately selected local extreme, not a representative estimate of ecological sensitivity.

is 65,661.6 proxy stems. The initial juvenile margins are 938.4 and 1,238.4 stems, and the basal-area threshold is 739.312 m², satisfied by both origins. At $t = 1$ and $\tau = 0$, Eq. (73) shows that the model continuation event is fully specified by terminal realization. The recorded viable fractions are 43/4096 and 3959/4096, a difference of 0.95605 under the selected conditional model comparison.

The path-resolved replay identifies the complete failure mechanism (Table 7). Both banks contain exclusively model-lawful histories. Each nonviable history violates the terminal juvenile

criterion; basal-area failures and joint juvenile/basal-area failures are absent, including under the unrestricted overlap census. The result therefore localizes the conditional capacity contrast to the juvenile-support continuation predicate. The requirement is $J_1 \geq 65,661.6$ stems. Pooled and

Terminal path outcome	Pooled exposure	Individual exposure
Model unlawfulness	0	0
Juvenile support failure only	4,053	137
Basal area failure only	0	0
Joint juvenile and basal-area failure	0	0
Successful lawful continuation	43	3,959
All candidate histories	4,096	4,096

Table 7: Executed mutually exclusive terminal failure census for the selected one-year conditional comparison. The separate overlapping-cause audit also finds zero basal-area failures and zero unlawful histories in either bank. All candidate histories retain their original denominator.

individual reconstructions begin at 66,600 and 66,900 juveniles; their respective reconstructed basal areas are 1,516.947 and 1,515.571 m², both well above the 739.312 m² basal-area threshold. Terminal juvenile outcomes straddle the declared requirement under the distinct conditional state distributions, while terminal basal-area adequacy persists in every simulated path. Thus the observed 3,916-history difference in continuing responses has a demonstrated predicate-level explanation within the model. This remains a deliberately selected local stress case and does not assign a unique ecological mechanism or pathwise causal identity to individual reconstructed stems.

6.5 Independent adult mortality: direct component-level evidence

The independently published HF453 survey supplies a support-aligned test of the earlier fitted adult hazard component. Of 21,578 eligible stems observed alive in 2021, 21,576 have a strict scored cumulative endpoint through 2024. No HF453 outcomes enter the fitted hazard distribution or select a calibration parameter. Table 8 compares the expected number of deaths from Eq. (103) with actual secure endpoint observations. Conditional cumulative mortality is underpredicted for the adult risk set, with the largest discrepancy concentrated in hemlock.

Risk set	n	Obs. deaths	Pred. deaths	Obs. rate	Pred. rate
All adult stems	21,576	1,477	1,050.285	0.06846	0.04868
Hemlock	7,557	685	280.348	0.09064	0.03710
Other taxa	14,019	792	769.938	0.05649	0.05492

Table 8: Independent adult-mortality component calibration in HF453 for the cumulative 2021-survivor to 2024 strict endpoint. Comparison respects adult size, stable tag linkage, observed 2021 survival, and secure status crosswalk; unobserved 2020–2021 outcomes are excluded. Predicted deaths use the earlier fitted hazard distribution and are not externally refitted.

Hemlock mortality has an observed-minus-predicted cumulative probability difference of 0.05355 (685 versus 280.348 deaths). The corresponding difference for nonhemlock adult stems is 0.00157 (792 versus 769.938). The adult survey also permits annual validation with changing risk sets (Table 9). In Table 9, O/P denotes observed/predicted event counts.

Interval	Adult n	All deaths O/P	Hemlock n	Hemlock deaths O/P	H. Brier
2021–22	21,576	449/361.47	7,557	183/95.05	0.02358
2022–23	21,124	516/350.98	7,372	226/91.83	0.02991
2023–24	20,608	509/339.75	7,146	274/88.33	0.03734

Table 9: Annual conditional mortality-component scores for independently observed adult stems, with each year’s observed risk set and secure-death classification. The cumulative 2024 totals are shown in Table 8; annual and cumulative sets have distinct eligibility and censoring.

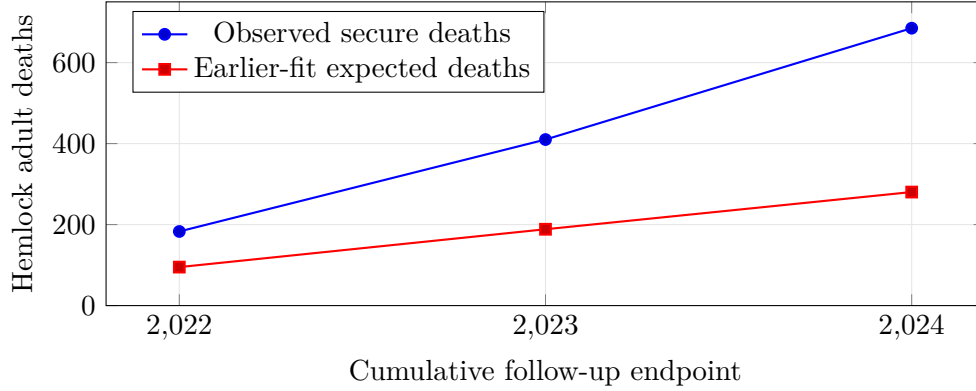


Figure 1: Independent HF453 cumulative adult hemlock mortality for a 2021-survivor, tagged-stem comparison. The later outcomes are held out from hazard fitting. Observed and predicted counts increase over follow-up under the separately recorded strict cumulative risk sets; the widening residual demonstrates persistent component-level miscalibration on this chosen support.

For cumulative 2024, a 1,000-replicate whole-quadrat spatial-cluster resampling gives a descriptive interval $[0.04557, 0.06175]$ for the hemlock observed-minus-predicted mortality-rate gap and $[-0.00358, 0.00725]$ for the other-taxon gap. The corresponding all-adult spatial-cluster interval is $[0.01501, 0.02472]$. These intervals condition on the selected observational support and resampling scheme; they do not claim coverage for an external ecological superpopulation.

The fitted cell-shared hazard distribution supplies a model-based predictive law distinct from the spatial-cluster descriptive intervals. Table 10 reports its independently evaluated median, central 95% equal-tail interval and upper-tail probability for each group on the same matched risk set. The observed hemlock mortality count of 685 lies far beyond the model’s predictive upper endpoint of 347; the computed probability of at least 685 deaths is 4.6308×10^{-21} . The 792 observed nonhemlock deaths fall within the interval 654–896 with upper-tail probability 0.3537. The model’s predictive variance separates event randomness conditional on the cell hazards from uncertainty in the shared hazards. The contributions are $266.318 + 782.132 = 1,048.450$ deaths² for hemlock, $693.360 + 3,121.328 = 3,814.688$ for nonhemlock and $959.678 + 3,903.460 = 4,863.138$ for all adults; the shared-hazard contributions comprise 74.60%, 81.82%, and 80.27% of the respective total variances. This uncertainty is included in Table 10. The 262,144 individual-event replicates yield zero hemlock exceedances, corresponding to a simulation-only 95% Wilson upper limit of approximately 1.4654×10^{-5} , while two agreeing positive deterministic probability calculations resolve the substantially smaller tail. Relative cross-method discrepancies are below 2.3×10^{-13} . The event simulations and independent conditional-count replicates agree with the central predicted distribution. The distinction between parameter uncertainty and individual biological frailty remains

Adult group	Observed	Mean	Median	Central 95%	$\Pr(D \geq D_{\text{obs}})$
Hemlock	685	280.348	279	220–347	4.6308×10^{-21}
Other taxa	792	769.938	768	654–896	0.353712
All adults	1,477	1,050.285	1,049	918–1,192	3.2045×10^{-8}

Table 10: Predictive mortality distribution under the stipulated independent fitted-cell parameter law, using the strict 2021-survivor to 2024 adult risk set. One hazard draw is shared across stems within each cell. The central intervals are model-based predictive intervals; the upper-tail probabilities are evaluated by independently checked positive-probability calculations and are distinct from empirical spatial-cluster bootstrap intervals.

explicit: the fitted law samples an uncertain cell-level hazard and contains no separately identified persistent stem-frailty distribution.

The strict cumulative 2024 risk set excludes two DN (dead/not found) endpoints. Treating these two as deaths gives 1,479 observed deaths among 21,578 stems, versus 1,050.371 expected; classifying both as alive gives 1,477 among 21,578, versus the same expected total. The primary strict score uses 1,477 among 21,576. The discrepancy remains under either recorded-category sensitivity. Starting-DBH groups show 1,213 versus 899.256 deaths for 10–30 cm stems ($n = 15,167$), and 264 versus 151.029 for stems of at least 30 cm ($n = 6,409$). Taxon, recorded-damage and protocol-proxy partitions, as well as prediction changes under ± 30 -day anniversary-time sensitivity, are separately preserved in the computational tables. The adult test is conditioned on observed survival to the 2021 baseline; unobserved deaths before that point and heterogeneity among individual tree hazards limit inference about the original 2020 prospective cohort. The concentrated hemlock discrepancy independently identifies component-level miscalibration within the chosen later survey support without identifying a unique causal pathogen or validating full forest Health.

The independent HF355 survey produces 8,840 eligible dated descriptive seedling survival transitions. Its observed population consists of below-1 cm seedlings sampled in small plots, with separately recorded first observation, survival and graduation constructs. The record does not identify whole-plot entry into the 1–10 cm proxy used by the Health requirements. This documents the observational support boundary of the available seedling evidence. Together, the adult and seedling datasets establish an independent component calibration evaluation and a separate nonidentification result, with their scientific targets distinguished.

6.6 Near-boundary decisions: observation, state and parameter attribution

The broad-range synthetic design contains 324 independently generated initial worlds, 69 true-positive and 255 true-negative Health cases in its reference correct-model experiment. Under the full-context information policy, the finite-design classifier obtains 99.22% balanced accuracy, with no primary true viable masses within 0.1 of its $\theta = 0.75$ boundary. Truth-generating omission of an annual hemlock-specific hazard 0.12 increases false-positive Health judgments from four to 34 and raises viable-mass mean absolute error from 0.01259 to 0.10705. All 324 signed mass errors in the hemlock-omission experiment are nonnegative; 275 are zero, so its mean signed bias and MAE are both +0.10705 in the reported orientation.

The deliberately boundary-concentrated roster contains 81 worlds, with 67 of the 72 capacity-targeted

worlds attaining $|\delta| \leq 0.10$ and 23 attaining $|\delta| \leq 0.02$. Certified presently realizing Q1 and Q2 cases near their margins exhibit balanced accuracies of 0.563 and 0.555, respectively. Their available Wilson and constructed-world resampling intervals are broad; reference-bank uncertainty persists in some cases after 65,536 sampled truths. The complete-roster context analysis uses 75 certified cases out of 81 for each query, including realization controls and cases outside the primary near-margin restriction. Its full-context balanced accuracy of 0.708 (Q1) and 0.652 (Q2) therefore estimates a different design-conditional quantity from the primary near-boundary summary.

To attribute this decision failure, the four-state/parameter information roles of Eq. (101) evaluate identical world targets and declared queries. Table 11 shows the certified presently realizing near-boundary outcomes from the nominal correct-kernel experiment.

Query	Information role	FP	FN	BA	Mass bias	Mass MAE	Certified n
Q1	Estimated state, estimated parameters	10	4	0.5630	-0.0889	0.2509	31
Q1	True state, estimated parameters	7	4	0.6513	-0.1361	0.2406	31
Q1	Estimated state, true parameters	8	7	0.5147	-0.1785	0.3677	31
Q1	True state, true parameters	0	0	1.0000	+0.00014	0.0040	31
Q2	Estimated state, estimated parameters	17	4	0.5549	-0.0256	0.2243	46
Q2	True state, estimated parameters	11	8	0.5890	-0.1150	0.2056	46
Q2	Estimated state, true parameters	15	7	0.5284	-0.0756	0.3171	46
Q2	True state, true parameters	0	0	1.0000	-0.000001	0.0038	46

Table 11: Stagewise oracle classification and continuous capacity errors on certified, presently realizing, $|\delta| \leq 0.1$ worlds of the nominal correct-model synthetic study. BA is balanced accuracy; bias is estimated minus true unconditional query mass. Continuous losses, signed bias and Boolean errors refer to the same certified evaluation supports. The joint oracle is an internal correct-kernel consistency control, not external forecast validation.

Query	Signed-margin region	Worlds/certified	Estimated-both MAE	Oracle-both MAE
Q1	Immediately above	9/6	0.2336	0.0046
Q1	Immediately below	9/6	0.2022	0.0037
Q1	Moderately above	10/10	0.2254	0.0043
Q1	Moderately below	9/9	0.3396	0.0049
Q2	Immediately above	12/9	0.2154	0.0043
Q2	Immediately below	11/8	0.1886	0.0035
Q2	Moderately above	13/13	0.2000	0.0038
Q2	Moderately below	10/10	0.3251	0.0045

Table 12: Continuous mass-error comparison by the registered signed-margin regions in the correct-kernel 81-world design. The paired information roles are estimated state/parameters (E00) and known state/parameters (E11). The first number gives all evaluated worlds in a region; the second gives worlds with certified reference truth for Boolean classification. The mass-error summaries use the complete region-specific numerical reference values and are conditional on the constructed roster.

True state improves some outcomes, while known parameters paired with an estimated state can worsen mass error. For Q2 the parameter-only improvement in absolute mass loss is -0.09281 , with a paired design-conditional interval $[-0.14574, -0.04111]$. The signed factorial interaction from Eq. (102) is -0.29460 , interval $[-0.34838, -0.23602]$. These interactions demonstrate that error

localization requires joint evaluation of state and parameter information, alongside finite predictive uncertainty. The oracle results are conditional on a constructed truth family; they make no claim of a uniquely dominant ecological cause. The source-qualified signed-margin analysis partitions the tested worlds by the true mass displacement $m - \theta$, separates presently realizing and nonrealizing states, and retains unresolved reference truths outside the certified confusion denominator. The reported paired uncertainty intervals refer to the constructed world’s specified resampling law; interactions cannot be read as an additive allocation of ecological error. A separate fixed-prefix numerical series (256, 1,024 and 4,096 futures) isolates predictive simulation variability from the four input-information interventions.

The reserve query’s single-interval inference leaves the original finite-bank judgments intact. In the shorter, separately identified 16,384-path reference study, nominal near-boundary Q2 balanced accuracy is approximately 0.552, with ten reference cases unresolved in the corresponding 53-case evaluation subset. These numbers describe that particular reference/selection resolution. They are not substituted for the larger reference’s certified truth table, and no 65,536-path result is attributed to the alternate simulation law.

6.7 Missingness, context, and finite-design decision sensitivity

The five specified latent-fate reconstruction families induce relatively small average finite-design Health disagreements, but their near-threshold effects are larger and more localized. Under the joint individual-time/mean-preserving-Gamma system and $\theta = 0.75$, the four alternatives have 24, 24, 39 and 42 semantic cells with present-near reversals against the baseline family. Local viable-mass differences can span the full unit interval. A separate diagnostic varies unknown-fate log-hazard shift by $-0.5, 0, 0.5$ and future candidate-entry multiplier by $0, 0.5, 1$, retaining each scenario/query combination. These are prescribed sensitivity coordinates, with no learned posterior frequency assigned to the mechanisms or entry weights.

Synthetic context labels supply hazard anchors (.005, .03, .12) and entry fractions (.025, .008, .001) to the inverse estimator. On the broad synthetic panel, balanced accuracy falls from 0.992 under full context to 0.919 after coarsening and 0.805 with both labels withheld. In the specially constructed boundary population, a coarser or unknown-context design mixture can outperform the full-label approximate estimator under its registered scoring policy. The boundary roster weights contexts unevenly; a separately specified roster-frequency prior is available and leaves the conditional interpretation unchanged. This is sensitivity of an approximate estimator and its design prior. Under a fixed joint law, target and loss, Bayes-optimal risk weakly decreases as the information sigma-field expands; the implemented comparisons do not constitute a counterexample to that principle.

The finite-count and uncertainty machinery also isolates numerical ambiguity. Q1 uses viable count and Q2 uses reserve-qualified count, each with its single component interval. Algebraic $\theta = 0$ and finite-bank $\theta = 1$ endpoints have their precise interpretations. A numerical certification of the stochastic generating law remains conditional on the declared model. No degree of path sampling by itself identifies biological entry or validates an externally misspecified hazard.

6.8 Query-relevant information and representation fidelity

The juvenile support query depends on individual diameters relative to 10 cm. The real E1 audit covers 53,611 securely observed living stems, in which cohort-RMS classification yields 1,187 false juvenile memberships and 18 missed juveniles, or 2.25% membership disagreement. Held-out masking also underestimates juvenile counts across the four declared families. These are observed-support comparisons, with current modeled origin juvenile counts still reconstructed from earlier observations.

Proposition 5.1 supplies the exact information result: count, RMS diameter and basal area do not determine all juvenile-sensitive queries. A size-aware two-component state retains additional threshold information with exact count and basal-area projection at the initial state. In the selected prospective comparison, only three of 1,536 finite-design Health cells differ, but one challenged state changes viable mass by $67/256 \approx 0.262$ under the enriched conditional implementation. The future stochastic generators change their arrays and have no established projection-commutation law; the numerical contrast is a conditional state/model experiment, while nonfactorization is exact.

The effective-count sensitivity experiment refits the hazard model under caps 50, 100, 200 and 500. Average 90% hazard-grid widths fall from about 0.0457 to 0.0312, while the largest mean absolute viable-mass change among the tested cap arms is 0.00322 and selected Health-label disagreement remains about 0.2% or less. These are bounded sensitivity findings, with the cap remaining a declared likelihood-tempering assumption. The model also includes a query-redundant deterministic density enrichment which preserves existing Q1/Q2 answers exactly. The richer size representation addresses a different information need and adds unresolved dynamic compatibility obligations.

7 Discussion

7.1 Identification and calibration as prerequisites for empirical capacity

The computational object studied here is a complete conditional measurement specification: censuses map to latent states; a parameter law and trajectory kernel generate unconditional candidate histories; model admissibility and continuation restrict those histories; and adequacy is evaluated on an unrenormalized viable subprobability measure. Each map is inspectable through equations, declared parameters, numerical counts and output provenance. This establishes a practical form of execution-level scientific accountability. Empirical use of its answers depends on what the measurements and dynamics can identify.

The independently scored adult-mortality component is particularly informative because it tests the earlier fitted law against different, later observations on a declared support. Its largest residual is hemlock-specific. That finding strengthens the controlled synthetic demonstration that omitted hemlock dynamics can inflate continuation mass and false-positive Health decisions. These results localize a model-stage weakness without assuming a unique species-specific ecological cause. Meanwhile, the seedling and tagged-stem identity analyses show that whole-plot entry into the juvenile proxy remains unidentified. Model precision cannot replace a missing biological identification variable. In particular, the 1.7 cm simulated birth assumption uses all recorded first-appearance size classes through Eq. (39); the candidate-size census constrains the raw count contribution of larger recorded stems, while the associated capacity consequences of a size-restricted refit have yet to be computed. This open sensitivity is retained as a specific scientific requirement

for interpreting regenerative and reserve probabilities.

The hemlock-specific predictive failure motivates a testable extension of the ecological state. Harvard Forest’s September 2025 third-census report records more than 11,800 dead stems in the 35-ha plot since 2020, including almost 5,000 eastern hemlocks attributed to hemlock woolly adelgid (*Adelges tsugae*) [19]; regional research associates infestation with foliar injury, crown decline and mortality [9]. These observations suggest that the fitted taxon–stratum–diameter hazard may omit evolving infestation pressure and accumulated organismal damage. A future study could enrich the state with serial crown-condition measurements, previously recorded insect-damage indicators and disturbance histories, and test a time-varying hazard $h_i(u \mid c(i), z_i(\leq u))$ against temporally held-out, support-matched tagged-stem mortality. Improved out-of-period calibration and reduced hemlock-specific predictive residuals would support this enrichment; persistent discrepancies would constrain the hypothesis. The third-census total covers a broader stem population and observation window than our 7,557 matched 2021-survivor adult hemlocks. It motivates this investigation while leaving causal attribution of the 685 observed matched deaths unresolved. The D2/D3 hemlock excess hazards remain stipulated scenario modifiers pending empirical identification of the disturbance process.

7.2 Decision relative validation and stage interactions

The broad-range and boundary-concentrated synthetic experiments pose distinct measurement questions. When generating worlds lie well apart from the adequacy boundary and the estimator receives informative, aligned context, its classifications are highly accurate. Near the threshold, marginal changes to estimated mass can change Boolean adequacy, and the joint inverse estimation problem exhibits materially weaker discrimination. The four-way oracle experiment demonstrates interactions between state and parameter information, including an adverse parameter-only intervention. The selected one-year exposure comparison demonstrates a large conditional capacity contrast whose failure mechanism is now numerically localized: every failed path in both banks violates the terminal juvenile-support requirement, while basal area and model admissibility remain satisfied. The independent adult-mortality test provides a complementary localization at the predictive-law stage. Its observed hemlock count lies beyond the model-based predictive distribution even after accounting for shared-hazard uncertainty, while nonhemlock mortality is compatible with that fitted law. These two results join logical predicate attribution with an external model-distribution test, while retaining the distinct inferential supports of each.

Additional numerical paths improve the approximation to a chosen stochastic kernel, while ecological calibration and scientific licensing concern the choice of the kernel and the organization itself. The distinct statuses of finite-bank truth, kernel uncertainty, reference truth ambiguity, source support, latent-state uncertainty and model discrepancy are therefore part of the measurement result. Decision-relative reporting makes these differences visible at the point where the declared adequacy language evaluates them.

7.3 Representation as a scientific information obligation

The RMS counterexample establishes that some response distinctions needed by a juvenile-sensitive query have been compressed from the retained cohort representation. A measured capacity summary and its adequacy query must be considered together: information sufficient for basal-area aggregation

can be insufficient for a thresholded juvenile-support requirement. Richer state summaries and independently observed size information are potential remedies, provided their observational ancestry and future compatibility are also evaluated. Initial count/second-moment projection by itself does not prove equality of stochastic future measures after a model change.

This distinction complements the deterministic, set-valued conservative-refinement guarantees established in Paper IV [4]. The present study concerns probabilities assigned by a declared stochastic generator to viable and failed histories. Equal attainable response support under two systems does not determine equal mass, and shared initial structural aggregates do not establish that two transition kernels commute under projection.

7.4 Boundaries of ecological authority and artifact reproducibility

The constitutive variables, four scenario modifiers, adequacy thresholds and protocol-derived wet-candidate stratum define a demonstrative organization. Empirical identification of its complete lawful and continuing history family would require additional biological state observations, a justified model of regeneration and species-specific dynamics, and independently evaluated future response laws. The adult-mortality score is a component-level retrospective test on a 2021-survivor-conditioned risk set. It integrates earlier-fit uncertainty about a shared cell hazard without fitting a persistent individual-frailty distribution or updating the hazard law for the 2021 selection event. The scored mean event probabilities and their support are explicit in Eq. (103); Eqs. (105)–(108) define the model-based predictive law and the executed external tail evaluation summarized in Table 10. Survivor-selection effects remain one possible source of differences between the fitted law and later observed mortality. The test was not a publicly issued prospective forecast. No result in this study assigns an operational ecological Health status to Harvard Forest.

The frozen computational study has completed independent, byte-exact reproduction in its declared native ARM reference environment. A private verification review inspected the returned execution evidence and confirmed all 24 prescribed scientific replay stages, all 69 returned evidence-file checksums, exact recovery of the original HFD-05 kernel fixture and HFD-06 fitted hazard posterior, and exact regeneration of the nine previously divergent predictive output files. The review rehased 9,683 stage-level frozen-file instances and 559 diagnostic-probe members without modifying the source, model inputs, expected numerical outputs, or acceptance criteria. The reference execution used Apple Silicon, Python 3.9.6, and NumPy 2.0.2; independent Intel execution exposed differences beginning at individual one-unit-in-the-last-place floating-point results. The numerical-platform comparison is retained as a portability qualification. These verified computational identities establish reproducibility of the stipulated procedure in its reference environment and leave its ecological validity subject to the independent empirical tests and limitations reported above.

The independently reviewed computational artifact is publicly released at https://github.com/zedjames/empirical_synthetic_forest_study, version v1.0.0, at the immutable scientific source commit a365fbffb6138a0c798276fe7b6f4577e9132305 [5]. The approved source snapshot provides the relevant scientific modules, configurations, documentation, and derived results; the two unchanged frozen ZIP assets provide the complete computational evidence, including 35 files too large for the browsable Git source subset. The public source review records dispositions for all 536 original candidate entries and the resolution of all 98 review-attention flags. Released original documentation and derived results carry the CC BY-NC-ND 4.0 terms, paper-specific executable code carries the PolyForm Noncommercial 1.0.0 terms, and third-party materials retain

their original conditions. The research repository remains private. Public readers can retrieve and audit the complete artifact under the declared native-ARM reference environment; the 24-stage exact scientific replay and its independent evidence inspection were performed before public distribution, and have not been represented as a newly executed third-party public reproduction. A separate computational-artifact DOI has not yet been assigned.

8 Conclusion

This study develops an inspectable stochastic implementation of prospective Health on a real, incomplete forest measurement substrate. The formal adequacy proposition is accompanied by explicit record-to-cohort, fitted-parameter, state-reconstruction, demographic trajectory and response-collection laws. Unconditional probability mass is retained through continuing and failing histories, with reserve adequacy reducible to a single mathematically defined restriction of viable mass.

Observed-data masking quantifies limitations of individual-diameter reconstruction. Stem identities and seedling support reveal a nonidentified whole-plot entry mechanism. Individual-time reconstruction exposes localized prospective-capacity sensitivity, including a selected one-year case in which all 4,190 nonviable histories across two conditional banks are assigned to terminal juvenile-support failure. Independent later adult observations identify substantial hemlock-specific predictive miscalibration, with 685 observed deaths exceeding the fitted model’s central interval of 220–347 and an upper-tail probability of 4.6308×10^{-21} after integrating shared-hazard uncertainty. Controlled boundary and oracle experiments establish the decision dependence on joint state and parameter information and on the distribution of adequacy margins. A finite counterexample establishes loss of query-required information under RMS compression. These results provide a stage-resolved account of achieved measurements together with the exact diagnostic observations needed to resolve remaining inferential distinctions.

The resulting work establishes an auditable computational measurement methodology under declared specifications, with identified successes, adverse tests and remaining scientific obligations. Its principal prospective Health outcomes retain their finite-design, model-conditional interpretation. Operational forest Health requires additional ecological identification, licensing and independent prospective validation.

A Executable study algorithms

A.1 Observation-to-state and finite capacity

1. Read exact versioned HF253 census bytes. Retain recorded identities/statuses, audit all eligible transitions, and apply the deterministic record-to-cell map in Eqs. (19)–(28).
2. Evaluate the unequal-interval hazard likelihood (33) on the declared 256-point grid and the joint parameter law (40). Preserve each pseudo-count/tempering and entry extrapolation assumption.
3. Sample reconstruction parameters and common latent shifts. For every actual dated component, draw its conditional survival indicator using Eq. (41); sample size and association branches and reaggregate as in Eq. (49). Retain measured-source and imputed-source provenance.
4. Independently draw predictive parameters and growth-model variant; iterate the annual generator with all four declared scenario modifiers. Store each candidate future and its original weight.
5. Test the exact model-admissibility and continuation predicates. Retain unsuccessful histories in the original candidate-law denominator and collect the viable subprobability capacity and complementary failure ledger.
6. Calculate the inclusive Q1/Q2 finite-bank Boolean and the separately reported pointwise stochastic-kernel MC status. Average conditional Boolean values across the explicit finite design measure, never thresholding an averaged capacity in its place.

A.2 Component validation and decision attribution

1. For masking calibration, generate predeclared masks from E0/campaign covariates, physically sanitize all held-out measurement fields, fit and reconstruct from the visible copy, and score the concealed observations only in the evaluation program.
2. For adult mortality, construct the 2021-survivor, earlier-adult, uniquely tagged HF453 risk set; enforce the status/date/size/exclusion definitions, compute Eq. (103) without fitting later outcomes, and reconstruct Brier/log/count summaries independently from raw records.
3. For synthetic truth, retain all designed initial worlds. Generate independent truth histories, create thinned/noisy observation packets, and run the strictly information-limited inverse estimator. Evaluate separated correct-kernel, hemlock-omission and entry-dispersion mechanisms.
4. For oracle attribution, score all four interfaces in Eq. (101); retain signed losses, margin strata, uncertified reference cases and interaction contrasts. Repeat bounded seed panels to expose finite future variance.
5. For sensitivity analysis, evaluate all declared reconstruction families, Gamma/exposure alternatives, cap values and size representations. Link each outcome to its exact model law, observational support, configuration and numerical bank. Retain all local extrema and unresolved cells.

B Computational provenance and audit scope

The computational program provides the following separate research objects and corresponding canonical source families. These are artifact identifiers for reproducibility, not manuscript draft versions.

Component	Computational record
Observed plot/source support	HFD01/object/primary_object.yaml; HFD01/epochs; HFD01/manifests/input_manifest.json
Fitted hazard and cohort generator	HFD02S/scripts/empirical.py; HFD02S/models/demography.py; HFD05/mathematical_audit/gamma_moments.md
Individual-time reconstruction	HFD05/exposure_model/individual_exposure_records.csv; cohort_exposure_inventory.csv; controlled_summary.csv
Q1/Q2 response arithmetic	HFD05/mathematical_audit/q2_equivalence.md; HFD05/q2_correction/resolution_comparison.csv
Corrected five-arm sensitivity	HFD05/corrected_pipeline/variant_matrix.csv; primary_surface.csv; missingness_surface.csv
Boundary and oracle results	HFD04/boundary_validation; HFD05/oracle_attribution/near_boundary_scores.csv; interactions.csv
External adult and seedling source	HFD05/external_sources/HF453/validation_summary.json; HF453/mortality_scores.csv; HF355/transition_summary.csv
Juvenile sufficiency and sensitivity	HFD04/juvenile_boundary/projection_counterexample.json; HFD05/corrected_pipeline/representation_surface.csv
Terminal path-predicate census	HFD06/attribution/partition.csv; HFD06/attribution/overlapping_causes.csv; HFD06/attribution/paths.csv
Shared-hazard predictive distribution	HFD06/mortality/final_predictive_summary.csv; HFD06/mortality/variance_decomposition.csv; HFD06/checks/rare_tail_precision.csv
Source/score verification	HFD05/reports/verification.json; mathematical_audit/pipeline_independent_audit.json; external_independent_audit.json

The completed output set includes exact finite-bank viable counts, matched adult expected-count scores, pathwise terminal failure classifications and a full shared-hazard mortality predictive distribution. The latter two tests reproduce the original fixed conditional state/parameter and matched adult risk-set identities, respectively, and have separately documented numerical checks. Appendix C records their executed identities and independent verification scope. The frozen packages use Python 3.9.6, NumPy 2.0.2, namespace-indexed PCG64 streams, fixed input identities, and output-file hashes. The private scientific closure certificate is preserved at verification-record commit 6b1bbe92c4cdcf64638ce97fc7e5b549c251c12 and issued on private main at ea3d151a2007049d0734ade84ecb918a0083dbd6. Its independent inspector verified 24 of 24 reference-ARM replay stages, 69 of 69 returned evidence checksums, 9,683 stage-level frozen-file comparisons, 559 probe-member comparisons, and nine of nine previously divergent HFD-06 numerical output identities. The original HFD-05 kernel fixture and HFD-06 hazard-posterior array were also reproduced at their frozen byte identities. The two operational bootstrap adjustments, the unsuccessful Intel replay, and the original negative scientific results remain separately preserved. No scientific model, random-stream definition, accepted tolerance, frozen file, or original input was altered in this closure. The reference runtime is native ARM64 on Darwin 27.0.0; the observed Intel deviations begin with one-ULP arithmetic differences, so cross-platform bitwise identity is not asserted.

The HFD-05 and HFD-06 scientific archive SHA-256 digests are respectively 45a10000799de1b7974e650d943adc7859428d830ab8e21f123506d92e715738 and 2a36eecd582a25db9b08e0fb258ddaaaa346e8dfea7809588cf751d3dcbc272. The returned ARM verification-evidence ZIP has SHA-256 135d06c780aaf3439b49e1946b888c3122f5d491b42b1079366df3a3daadf1ae and is preserved in an access-controlled private artifact location outside ordinary Git history. The independently inspected original HFD-05 and HFD-06 ZIPs are published without byte modification as the two assets of v1.0.0; their SHA-256 digests above agree with the public SHA256SUMS manifest. The public scientific source is pinned by commit a365fbff6138a0c798276fe7b6f4577e9132305, the target of release tag v1.0.0 [5]. The repository records a 536-entry disclosure inventory, the reviewed published source subset, and exact reproduction instructions; the full scientific replay is conducted from the complete ZIPs under the reference runtime. Independent scientific verification occurred on the isolated original packages before publication, while public artifact distribution was separately checked against the frozen ZIP identities. No separate Zenodo DOI for this computational artifact is asserted. The privately held verification evidence remains internal provenance, and the chronology of the private repository is not characterized as third-party preregistration.

C Terminal evidence and independent numerical checks

The selected one-year path-stage study uses state 205, parameter replicate 1, D0, Gamma growth, $(\alpha, \beta, \tau) = (0.5, 0.8, 0)$, and two separately reconstructed initial states. The complete 8,192 simulated histories reproduce the 43 and 3,959 lawful-and-continuing outcomes in the registered 4,096-history banks. Each history is classified independently from its terminal (J_1, BA_1) and model-lawful indicator by Eq. (75). The exclusive results satisfy

$$\begin{aligned} 4096 &= 0 + 4053 + 0 + 0 + 43, \\ 4096 &= 0 + 137 + 0 + 0 + 3959, \end{aligned} \tag{117}$$

where each ordered tuple corresponds to unlawful, juvenile-only failure, basal-area-only failure, joint juvenile/basal-area failure, and continuing histories. Independently computed overlapping flags confirm the absence of basal-area and unlawful failures in either bank. An independent scalar reconstruction checks original and terminal trajectory records and the full indicator census. The high-precision selection is a local diagnostic under the declared finite design, with separate future laws conditioned on different reconstructed origins.

The external predictive assessment uses the fixed 21,576-adult strict cumulative 2024 matched risk set, including 7,557 hemlock and 14,019 other adult stems, and the previously fitted 256-point cell-hazard grids. The predictive sampler draws one hazard per occupied cell and conditionally generates individual Bernoulli mortality outcomes. Across 262,144 complete predictive replicates it generates 5,656,018,944 individual events, while a separately implemented cell-conditional Binomial-count sampler generates another 262,144 replicates per adult grouping for an independent central-distribution check. In the latter, the strict cumulative endpoint assigns $\Delta_i = 3$ years for every included stem; hence all stems in an occupied original hazard cell share $q_c(H_c) = 1 - e^{-3H_c}$ and the cell death count is Binomial conditional on the single sampled H_c . Group totals sum independent cell counts under the declared across-cell parameter law. This exact equivalence applies to the executed fixed-anniversary predictive check; it is not asserted for unequal individual exposures. The model-based mean/variance/cumulative probabilities are also calculated from the full discrete fitted cell laws and checked by independent positive convolution methods. At extreme upper-tail

events, simulation-based zero-hit frequencies are distinguished from positive deterministic probability calculations. The hemlock upper-tail probability 4.630784×10^{-21} is verified by two separate positive numerical routes, with relative discrepancy below 2.3×10^{-13} . Across both terminal studies, twelve active hostile controls and four isolated replay stages pass. These are conditional simulation and numerical-verification statements; access for independent inspection remains subject to publication of the reviewed computational artifact.

D Scientific authority and series boundary

Papers I–III provide the constitutive definition, contextual licensing and query-relative representation theory [1, 2, 3]; Paper IV supplies an exact finite semantic-provenance and conservative-refinement benchmark [4]. The present empirical-synthetic study concerns the stochastic observation-to-capacity calculation and the measurement evidence for its individual stages. The exact deterministic set-valued refinement theorem fixes a shared dynamics/constitution and exact capacity projection. Its conclusion does not extend automatically to independently changed candidate-history measures or demographic generators. The adult component test, seedling survey and synthetic oracle experiments provide distinct and appropriately limited empirical and designed-world evidence. Independent future issuance of an operational ecological Health assessment would require a harmonized baseline, ecologically licensed constitution and prospective law, and an archived forecast before access to evaluation outcomes.

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